

MODULE CODE: AP80167
MODULE TITLE: PHYSICS FOR IT

BY

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University of Rwanda, CST, LEVEL 1

Assessment Pattern

Component	Weighting(%)
Assignments and Lab repots	10
Quizzes	10
One Continuous Assessment Test(CAT) in the middle of the module	30
Final written Exam	50

Indicative Content

Part 1: **Electricity**

Part 2: **Magnetism**

ELECTRICITY



0. Course outline

- **Electric forces and electric fields:** Properties of electric charges, Insulation and conductors, electrification; Coulomb's law, the electric field, Electric field lines, Electric flux.
- **Electric energy and capacitance:** Electric potential, Potential energy due to point charges, equipotential surfaces, Capacitance, Combination of capacitors, Energy stored in a charged capacitor, dielectrics.
- **Current and resistance:** Electric current, Ohm's law, Resistivity and resistance, Electromotive force, Generators, Electric circuit, Resistors in series, Resistors in parallel, Kirchhoff's laws, Electrical energy and power, Electrical safety

1. Introduction

Electricity is a general term encompassing a variety of phenomena resulting from *the presence and flow of electric charges*.

These include many easily recognizable phenomena, such as lightning, static electricity, and the flow of electrical current in electrical wires.

In addition, electricity encompasses (includes) less familiar concepts such as the electromagnetic field and electromagnetic induction.

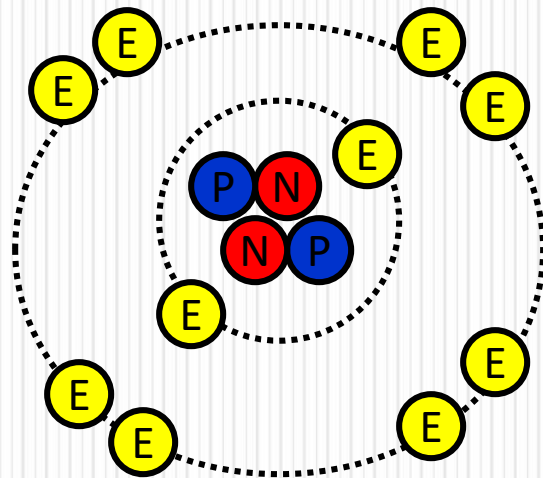
2. Electric charge

Charge originates in the atom, in which its most familiar carriers are the **electron** and **proton**.

2. Electric charge

• Atom Parts:

- **Electrons:** Negative part of an atom.
- **Protons:** Positive part of an atom.
- **Neutrons:** No charged part of an atom.



2. Electric charge

There are two types of observed electric charge: **positive** and **negative**.

- The convention was derived from Benjamin Franklin's experiments.
- Charge with opposite electrical sign (i.e. a positive charge and a negative charge) attract each other and charge with the same electrical sign (i.e. two positive charges or two negative charges) repel each other.

2. Electric charge

The smallest unit of “free” charge known in nature is the charge of an electron or proton, which has a magnitude of

$$e = 1.602 \times 10^{-19} \text{ C}$$

Charge comes in multiples of an indivisible unit of charge, represented by the letter e .

These things have the same size charge, but the sign is different.

A proton has a charge of $+e$, while an electron has a charge of $-e$.

2. Electric charge

- Putting "charge is quantified" in terms of an equation, we say:

$$q = n e$$

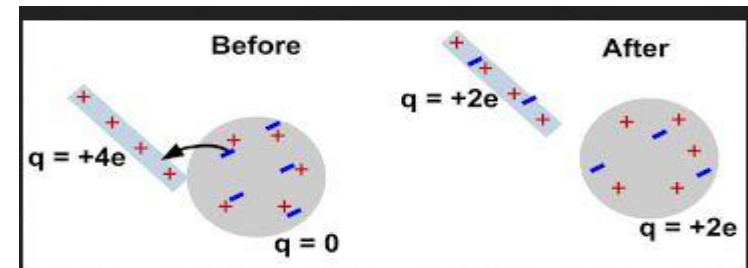
- where q is the symbol used to represent charge, while n is a positive or negative integer, and e is the electronic charge, $1.60 \times 10^{-19} \text{ C}$.

2.1. Conservation of Charge

The Law of conservation of charge states that “*The net charge of an isolated system remains constant.*”

If a system starts out with an equal number of positive and negative charges, there is nothing we can do to create an excess of one kind of charge in that system unless we bring in charge from outside the system (or remove some charge from the system).

Likewise, if something starts out with a certain net charge, say $+100e$, it will always have $+100e$ unless it is allowed to interact with something external to it.



2.1. Conservation of Charge

- Charge can be created and destroyed, but only in positive-negative pairs.
- Within the system, charge may be transferred between bodies, either **by *direct contact*, or by *passing along a conducting material*, such as a wire.**
- The informal term **static electricity** refers to **the net presence (or 'imbalance') of charge** on a body, usually caused when dissimilar materials are rubbed together, transferring charge from one to the other.

2.2. Electrostatic charging

- Forces between two electrically-charged objects can be extremely large.
- Most things are electrically neutral; they have equal amounts of positive and negative charge. If this wasn't the case, the world we live in would be a much stranger place.
- We also have a lot of control over how things get charged. This is because we can choose the appropriate material to use in a given situation.

2.2. Electrostatic charging

- Metals are *good conductors* of electric charge, while **plastics, wood, and rubber** are not. They are called *insulators*.
- Charge does not flow easily through insulators as it does through conductors, which is why wires you plug into a wall socket are covered with a protective rubber coating. Charge flows along the wire, but not through the coating to you.

2.2. Electrostatic charging

Materials are divided into three categories, depending on how easily they will allow charge (i.e., electrons) to flow along them. These are:

- *conductors* - metals, for example
- *semi-conductors* - silicon is a good example
- *insulators* - rubber, wood, plastic for example

Most materials are either conductors or insulators.

- The difference between them is that in conductors, the outermost electrons in the atoms are so loosely bound to their atoms that they are free to travel around.
- In insulators, on the other hand, the electrons are much more tightly bound to the atoms, and are not free to flow. Semi-conductors are a very useful intermediate class, not as conductive as metals but considerably more conductive than insulators.
- By adding certain impurities to semi-conductors in the appropriate concentrations the conductivity can be well-controlled.

Energy Band Diagrams

— ***Valence Electrons*** are those electrons which are located in the outermost or “**Valence**” shell of an atom.

— The number of valence electrons an atom has determines the electrical properties of that atom.

< 4 electrons => Conductor

> 4 electrons => Insulator

4 electrons => Semiconductor

2.2. Electrostatic charging

There are three ways that objects can be given a net charge. These are:

a) ***Charging by friction*** - this is useful for charging insulators.

If you rub one material with another (say, a plastic ruler with a piece of paper towel), electrons have a tendency to be transferred from one material to the other.

2.2. Electrostatic charging

b) Charging by contact - useful for charging metals and other conductors.

If a charged object touches a conductor, some charge will be transferred between the object and the conductor, charging the conductor with the same sign as the charge on the object.

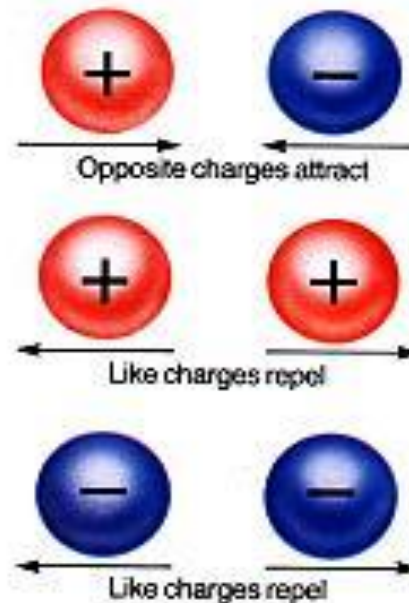
2.2. Electrostatic charging

c) Charging by induction - also useful for charging metals and other conductors.

Again, a charged object is used, but this time it is only brought close to the conductor, and does not touch it.

3. Coulomb's law (inter-action between charges)

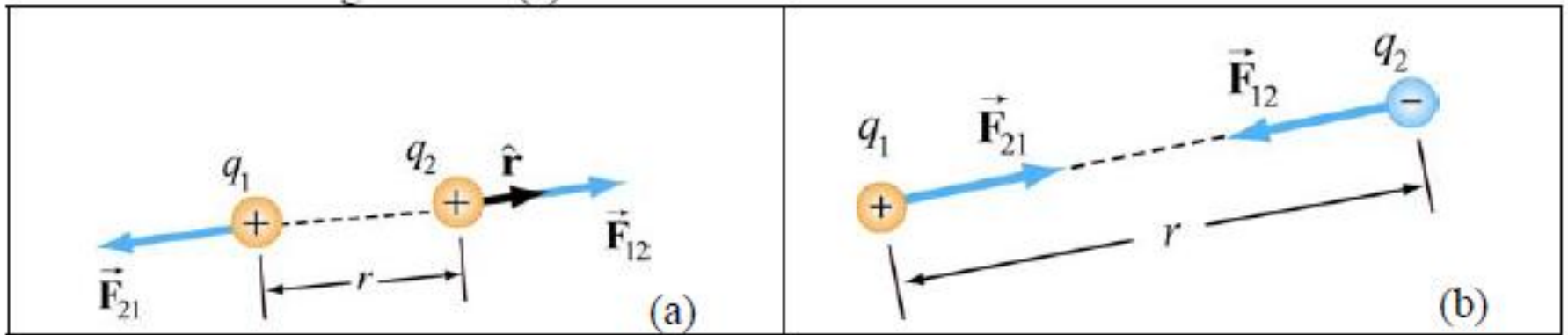
Two electric charges of the same sign repel and two electric charges of opposite sign attract each other with a force that is proportional to the product of their charges and inversely proportional to the square of the distance between them.



3. Coulomb's law

If we consider a system of two charges separated by a distance r , the force between those two charges is given by

$$\vec{F}_{21} = k \frac{q_1 q_2}{r^2} \hat{r}$$



$$\hat{r} = \frac{\vec{r}}{r}$$

3. Coulomb's law

- Note that electric force is a vector which has both magnitude and direction.
- In SI units, the Coulomb's constant k is given by

$$k = \frac{1}{4\pi\epsilon_0} = 8.9875 \times 10^9 \text{ Nm}^2/\text{C}^2$$

where ϵ_0 is permittivity of free space

$$\epsilon_0 = \frac{1}{4\pi k} = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N} \cdot \text{m}^2}$$

3.1. Principle of superposition

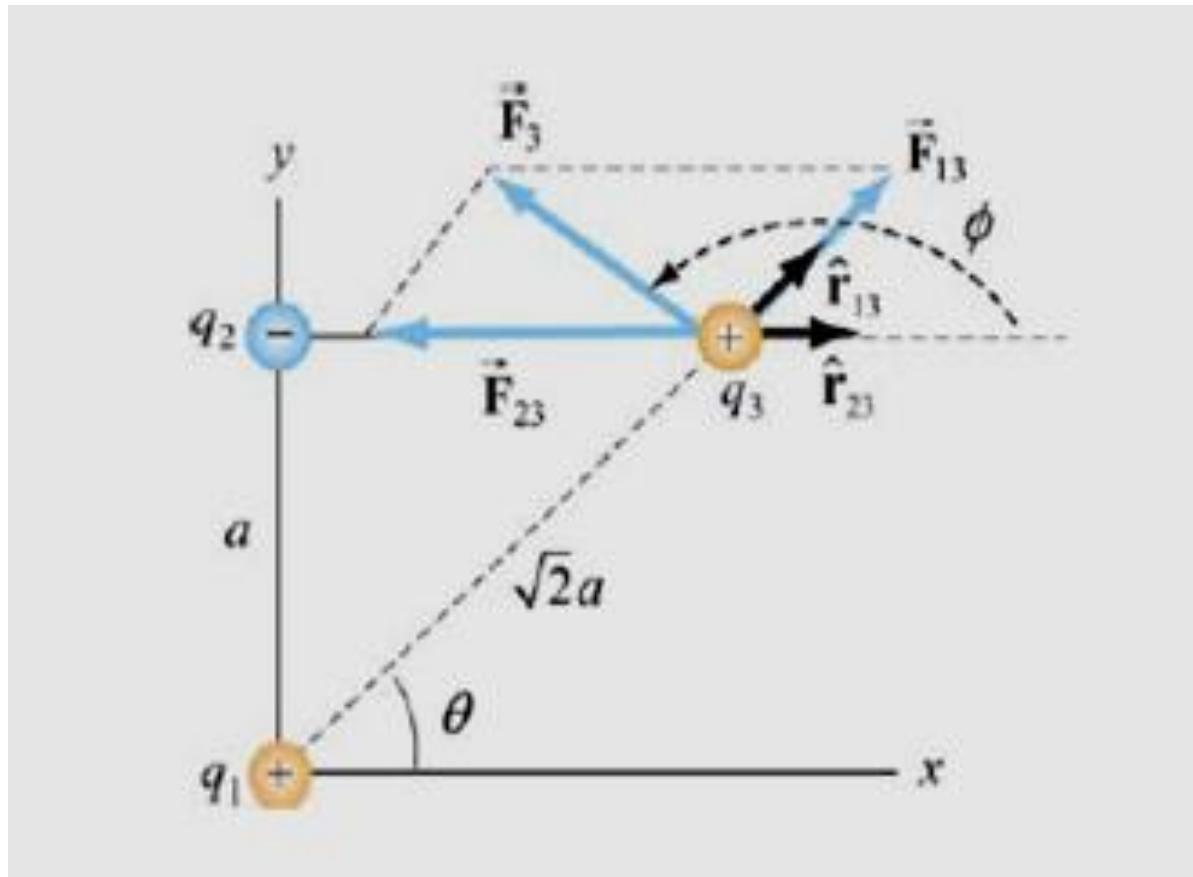
Coulomb's law applies to any pair of point charges.

When more than two charges are present, the net force on any one charge is simply the vector sum of the forces exerted on it by the other charges.

For example, if three charges q_1 , q_2 and q_3 are present, the resultant force experienced by q_3 due to q_1 and q_2 will be

$$\vec{F}_3 = \vec{F}_{13} + \vec{F}_{23}$$

3.1. Principle of superposition



$$\vec{F}_3 = \vec{F}_{13} + \vec{F}_{23}$$

3.1. Principle of superposition

- For a system of N charges, the net force experienced by the j^{th} particle would be

$$\vec{F}_j = \sum_{\substack{i=1 \\ i \neq j}}^N \vec{F}_{ij}$$

Where $\vec{F}_{ij}$ denotes the force between particles i and j .

- The superposition principle implies that the net force between any two charges is independent of the presence of other charges, This is true if the charges are in fixed positions.*

4. Electric Field

The region around a charged body within which it can exert its electrostatic influence may be called an *electric field*.

In principle, it extends to infinity, but in practice it falls off more or less rapidly with distance.

We can define the *intensity* or *strength* E of an electric field as follows: Suppose that we place a small test charge $+q$ in an electric field. This charge will then experience a force.

The ratio of the force to the charge is called the *intensity of the electric field*, or, more usually, simply the *electric field*.

Thus we have used the words “electric field” to mean either the region of space around a charged body, or, quantitatively, to mean its intensity.

Usually it is clear from the context which is meant, but, if you wish, you may elect to use the longer phrase “intensity of the electric field” if you want to remove all doubt.

- *The field and the force are in the same direction*, and the electric field is a vector quantity, so the definition of the electric field can be written as

$$\vec{E} = \frac{\vec{F}_e}{q_0}$$

- We take q_0 to be infinitesimally small so that the field generated does not disturb the “*source charges*.”

- Using the definition of electric field $\vec{E} = \frac{\vec{F}_e}{q_o}$ and the Coulomb's law, the electric field at a distance r from a point charge q is given by

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

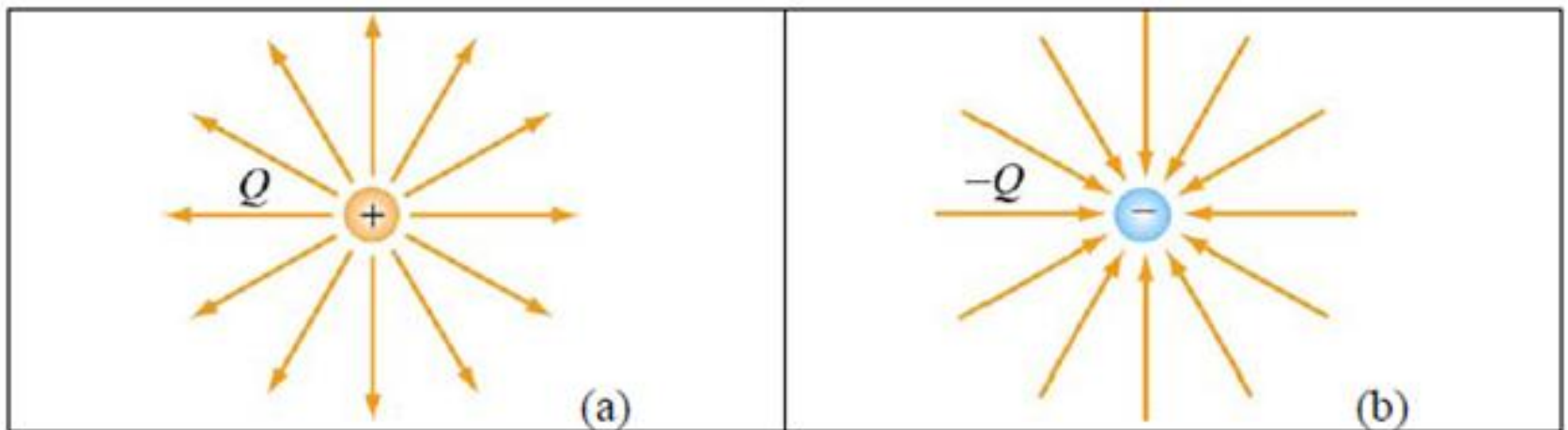
- Using the superposition principle, the total electric field due to a group of charges is equal to the vector sum of the electric fields of individual charges:

$$\vec{E} = \sum_i \vec{E}_i = \sum_i \frac{1}{4\pi\epsilon_0} \frac{q_i}{r_i^2} \hat{r}_i$$

4.1. Electric Field Lines

Electric field lines provide a convenient graphical representation of the electric field in space.

The field lines for a positive and a negative charges are shown in figure below



4.1. Electric Field Lines

Notice that the direction of field lines is *radially outward for a positive charge and radially inward for a negative charge*.

For a pair of charges of equal magnitude but opposite sign (an electric dipole), the field lines are shown in figure below:



4.1. Electric Field Lines

The pattern of electric field lines can be obtained by considering the following:

- a) Symmetry:* For every point above the line joining the two charges there is an equivalent point below it. Therefore, the pattern must be symmetrical about the line joining the two charges;
- b) Near field:* Very close to a charge, the field due to that charge predominates. Therefore, the lines are radial and spherically symmetric;

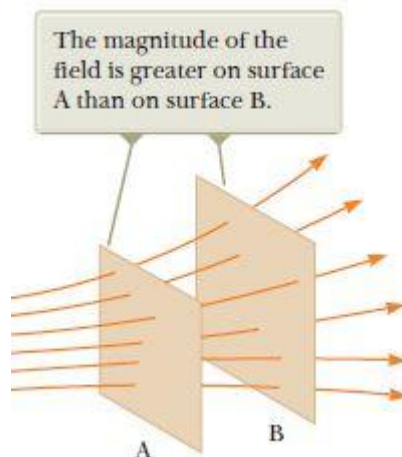
4.1. Electric Field Lines

- c) Far field:* Far from the system of charges, the pattern should look like that of a single point charge of value $q = \sum_i q_i$. Thus, the lines should be radially outward, unless $Q=0$;
- d) Null point:* This is a point at which $\vec{E} = \mathbf{0}$, and no field lines should pass through it.

4.1. Electric Field Lines

The properties of electric field lines may be summarized as follows:

- ✓ The direction of the electric field vector at a point is tangent to the field lines;
- ✓ The number of lines per unit area through a surface perpendicular to the line is devised to be proportional to the magnitude of the electric field in a given region;
- ✓ The field lines must begin on positive charges (or at infinity) and then terminate on negative charges (or at infinity);

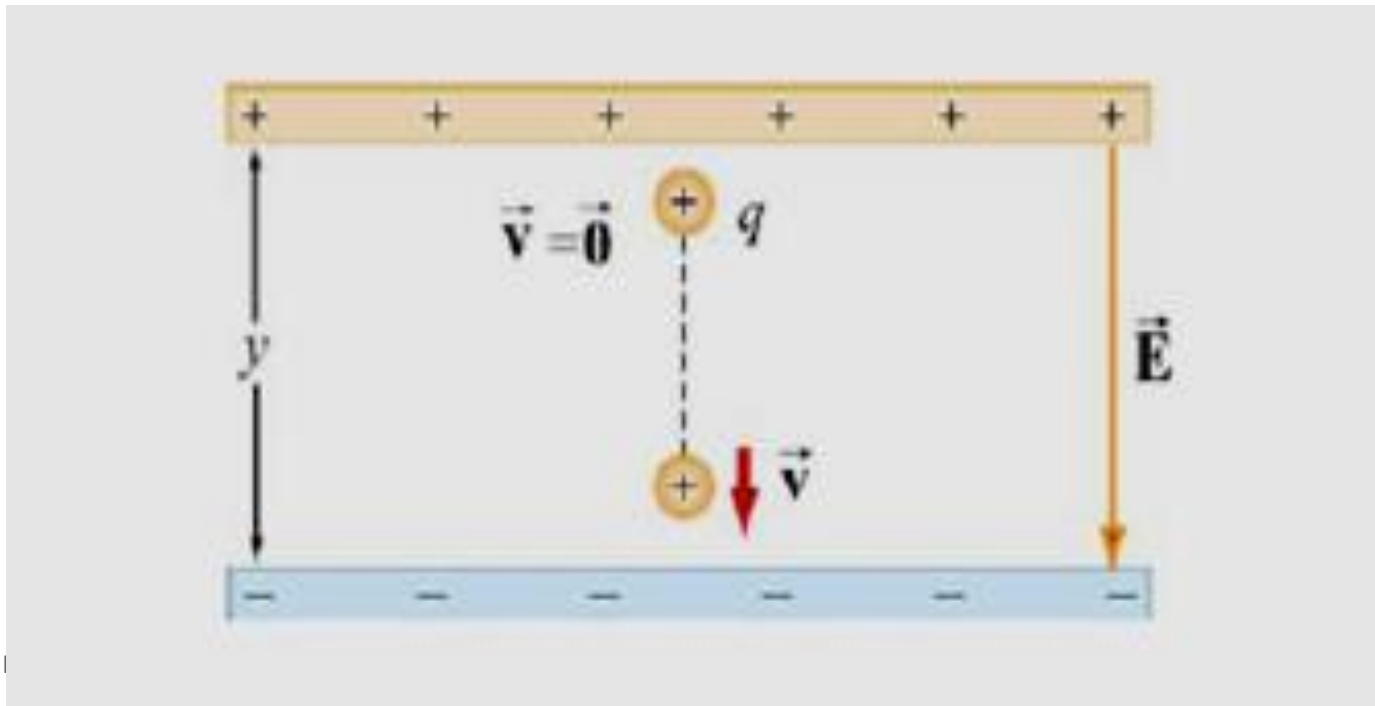


4.1. Electric Field Lines

- ✓ The number of lines that originate from a positive charge or terminating on a negative charge must be proportional to the magnitude of the charge;
- ✓ No two field lines can cross each other; otherwise the field would be pointing in two different directions at the same point pass through it.

4.1. Force on a Charged Particle in an Electric Field

Consider a charge $+q$ moving between two parallel plates of opposite charges, as shown in Figure below:



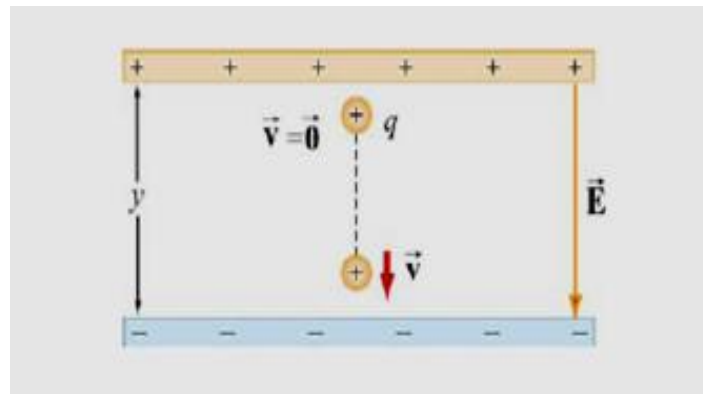
4.1. Force on a Charged Particle in an Electric Field

Let the electric field between the plates be $\vec{E} = E_y \hat{j}$

The charge will experience a downward Coulomb force $\vec{F}_e = q\vec{E}$

According to Newton's second law, a net force will cause the charge to accelerate with an acceleration,

$$\vec{a} = \frac{\vec{F}_e}{m} = \frac{qE_y}{m} \hat{j}$$



4.1. Force on a Charged Particle in an Electric Field

Suppose the particle is at rest ($v=0$) when it is first released from the positive plate. The final speed v of the particle as it strikes the negative plate is given by

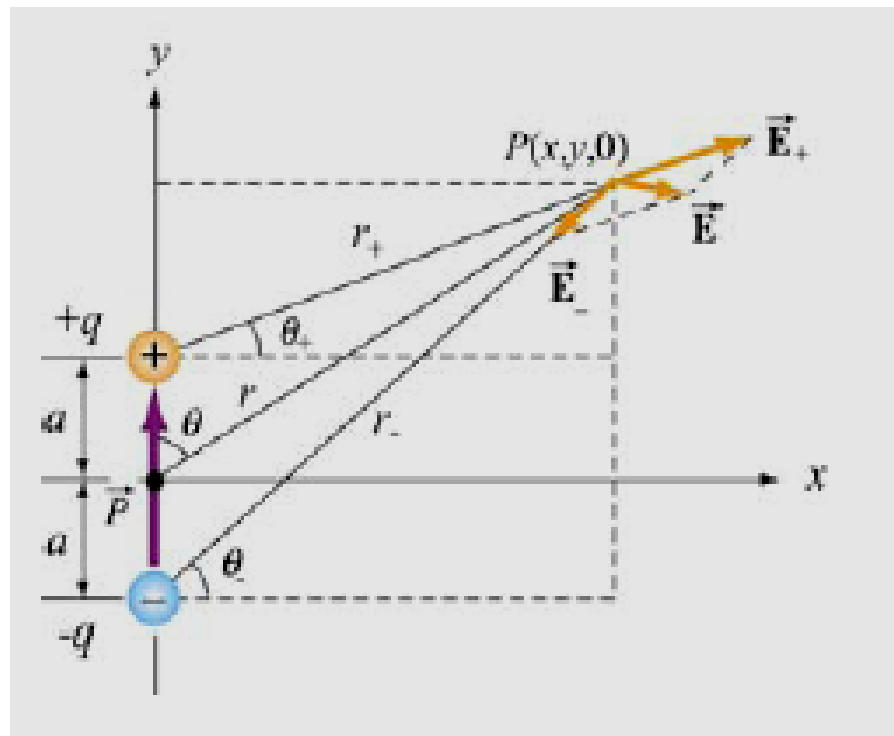
$$v_y = \sqrt{2|a_y|y} = \sqrt{\frac{2yqE_y}{m}}$$

where y is the distance between the two plates. The kinetic energy of the particle when it strikes the plate is given by

$$K = \frac{1}{2}mv_y^2 = qE_y y$$

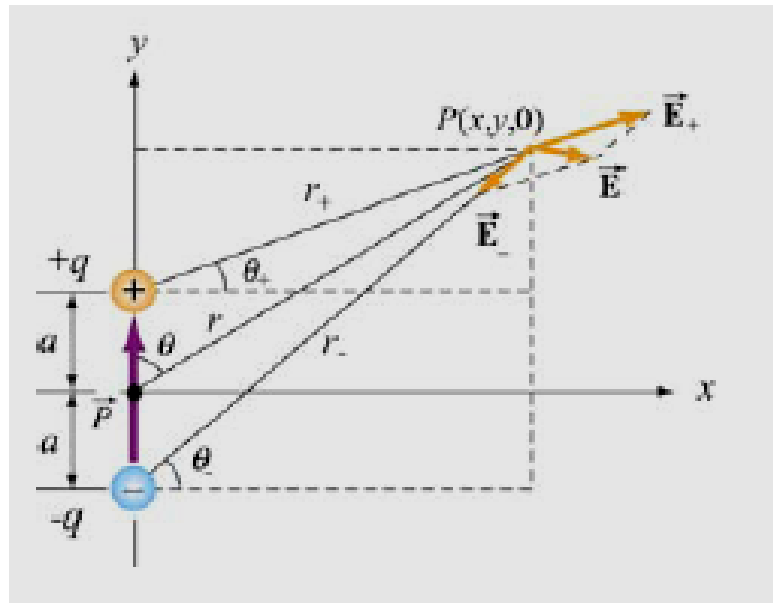
4.3. Electric Dipole

An electric dipole consists of two equal but opposite charges, $+q$ and $-q$, separated by a distance $2a$, as shown in the figure below:



4.3. Electric Dipole

The dipole moment vector $\vec{p}$ which points from $-q$ to $+q$ (in the y -direction) is given by $\vec{p} = 2qa\hat{j}$



4.3. Electric Dipole

The magnitude of the electric dipole is $p = 2qa$

with $q > 0$

For an overall charge-neutral system having N charges, the electric dipole vector is defined as $\vec{p}$

$$\vec{p} = \sum_i^N q_i \vec{r}_i$$

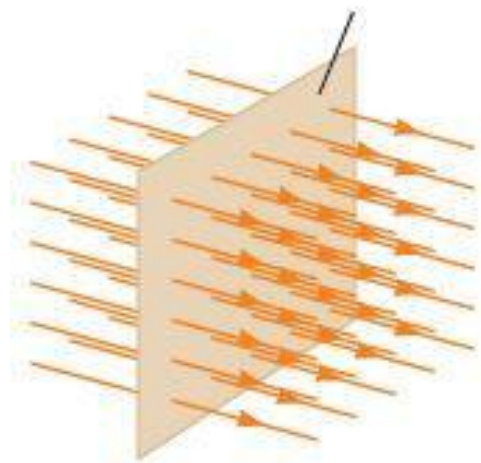
Where $\vec{r}_i$ is the position vector of the charge q_i

4.3. Electric Dipole

An electric dipole is a separation of positive and negative charges. The simplest example of this is a pair **of electric** charges of equal magnitude but opposite sign, separated by some (usually small) distance. A permanent **electric dipole** is called an electret.

Examples of dipoles include HCl, CO, H₂O and other *polar* molecules. In principle, any molecule in which the centers of the positive and negative charges do not coincide may be approximated as a dipole.

Field lines representing a uniform electric field penetrating a plane of area perpendicular to the field



Consider an electric field that is uniform in both magnitude and direction as shown in Figure at left side. The field lines penetrate a rectangular surface of area whose plane is oriented perpendicular to the field. The number of lines per unit area (in other words, the *line density*) is always proportional to the magnitude of the electric field. Therefore, the total number of lines penetrating the surface is proportional to the product EA . This product of the magnitude of the electric field and surface area perpendicular to the field is called the **electric flux**. From the SI units of E and A , we see that ϕ_E has units of Newton meters squared per coulomb $\mathbf{N \cdot m^2 / c}$

4.5. Electric flux

If the surface under consideration is not perpendicular to the field, the flux through it must be less than that given by the equation above.

To calculate the flux through a particular surface, multiply the surface area by the component of the electric field perpendicular to the surface.

$$\Phi = \vec{E} \cdot \vec{A} = EA \cos \varphi$$

From this result, we see that the flux through a surface of fixed area A has a maximum value EA when the surface is perpendicular to the field (when the normal to the surface is parallel to the field, that is, when $\varphi = 0^\circ$); the flux is zero when the surface is parallel to the field (when the normal to the surface is perpendicular to the field, that is, when $\varphi = 90^\circ$).

Where φ is the angle between the electric field and the normal to the surface

4.6. Gauss's law

Gauss's law states that *“the flux of the electric field over the Gaussian surface equals to the net charge enclosed by that surface.”*

$$\Phi = \frac{q}{\epsilon_0}$$

- The integral form of Gauss's law:

$$\int \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$$

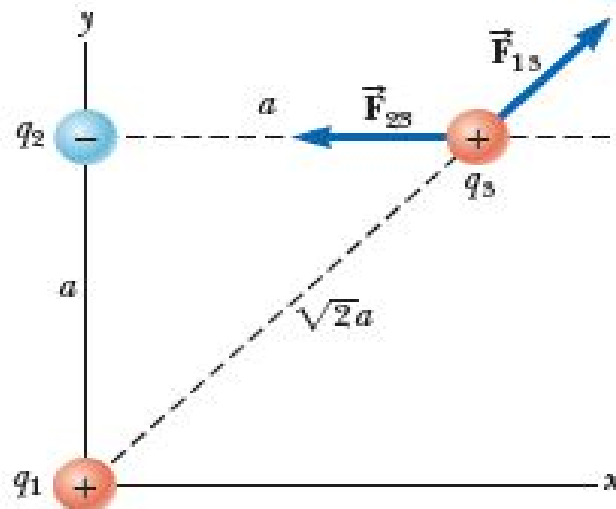
Exercise 1

The electron and proton of a hydrogen atom are separated (on the average) by a distance of approximately $5.3 \times 10^{-11} \text{ m}$.

Find the magnitudes of the electric force and the gravitational force between the two particles.

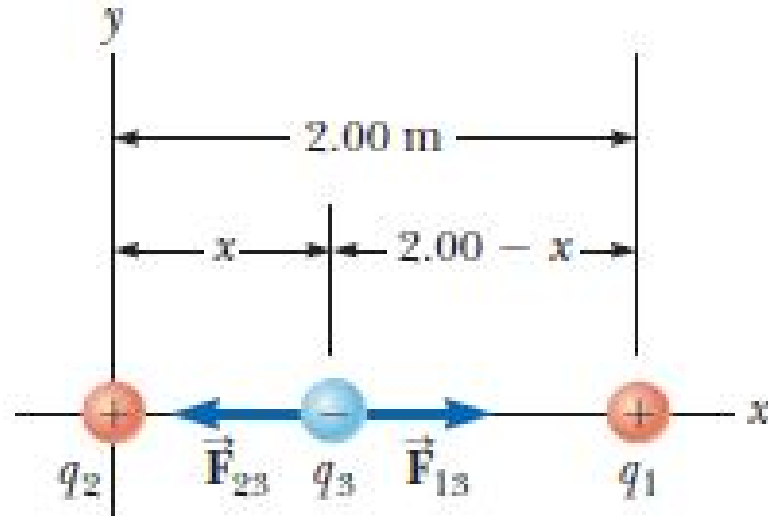
Exercise 2

Consider three point charges located at the corners of a right triangle as shown in Figure below, where $q_1 = q_3 = 5.00\mu\text{C}$, $q_2 = -2.00\mu\text{C}$ and $a=0.100\text{m}$. Find the resultant force exerted on q_3 .



Exercise 3

Three point charges lie along the x axis as shown in Figure below. The positive charge $q_1 = 15.0\mu\text{C}$ is at $x = 2.00\text{m}$, the positive charge $q_2 = 6.00\mu\text{C}$ is at the origin, and the net force acting on q_3 is zero. What is the x coordinate of q_3 ?



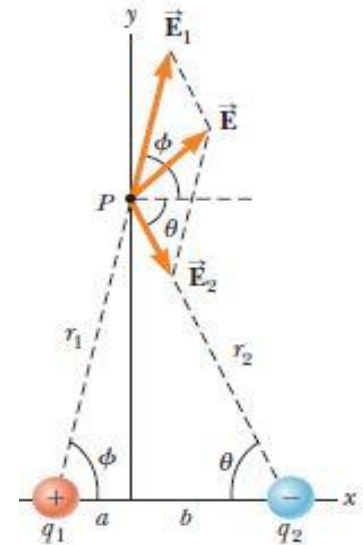
Exercise 4

A water droplet of mass 3.00×10^{-12} Kg is located in the air near the ground during a stormy day. An atmospheric electric field of magnitude 6.00×10^3 N/C points vertically downward in the vicinity of the water droplet. The droplet remains suspended at rest in the air. What is the electric charge on the droplet?

Exercise 5

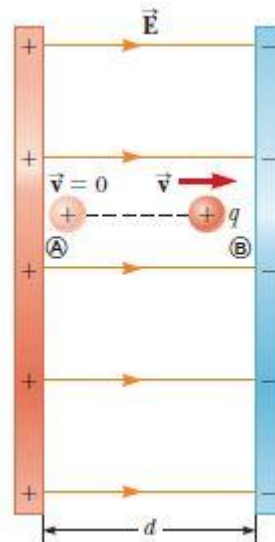
Charges q_1 and q_2 are located on the x axis, at distances a and b , respectively, from the origin as shown in Figure below.

- (A) Find the components of the net electric field at the point P , which is at position $(0, y)$.
- (B) Evaluate the electric field at point P in the special case that $|q_1| = |q_2|$ and $a = b$.
- (C) Find the electric field due to the electric dipole when point P is a distance $y \gg a$ from the origin.



Exercise 6

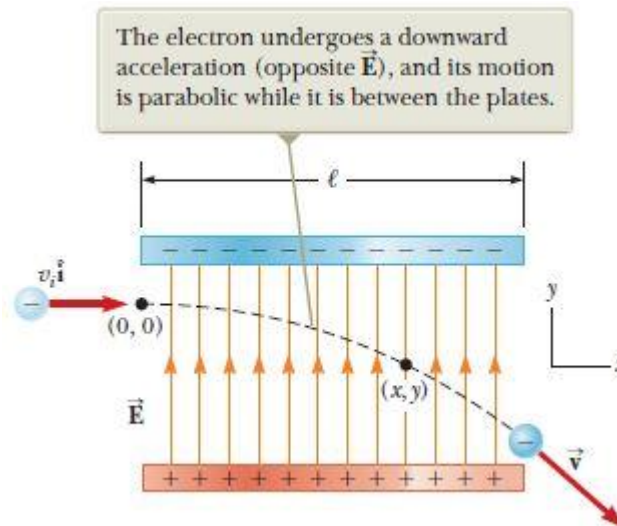
A uniform electric field $\vec{E}$ is directed along the x axis between parallel plates of charge separated by a distance d as shown in Figure. A positive point charge q of mass m is released from rest at a point **A** next to the positive plate and accelerates to a point **B** next to the negative plate. Find the speed of the particle at **B** by modeling it as a particle under constant acceleration.



Exercise 7

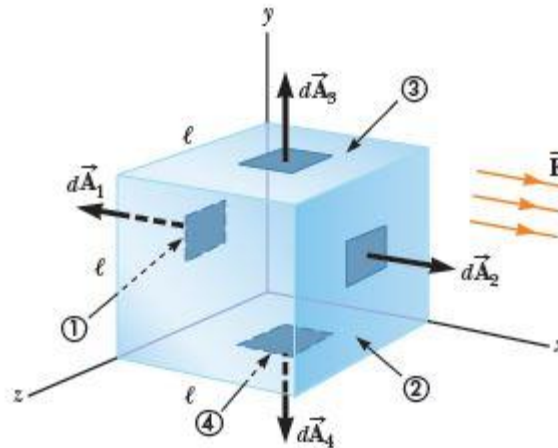
An electron enters the region of a uniform electric field as shown in Figure below, with $v_i = 3.00 \times 10^6$ m/s and $E = 200$ N/C. The horizontal length of the plates is $l = 0.100$ m.

- (A) Find the acceleration of the electron while it is in the electric field.
- (B) Assuming the electron enters the field at time $t = 0$, find the time at which it leaves the field.
- (C) Assuming the vertical position of the electron as it enters the field is $y_i = 0$, what is its vertical position when it leaves the field?



Exercise 8

Consider a uniform electric field $\vec{E}$ oriented in the x direction in empty space. A cube of edge length ℓ is placed in the field, oriented as shown in Figure below. Find the net electric flux through the surface of the cube.



QUIZ 1

5. Electric Potential or Voltage

The best way to think about the meaning of *electric potential*, or *voltage*, is to make an analogy with gravity.

When a mass is high above the ground we say that it has a lot of potential energy because if we drop it, it strikes the ground hard.

We say that the potential energy due to height has been converted into kinetic energy. We can say the same things about electricity.

5. Electric Potential or Voltage

If two positive charges are close together and we release them, they will repel each other and pick up speed, their electric potential energy having been converted into kinetic energy.

In this analogy, the electrical quantity which corresponds to height is **voltage**.

5. Electric Potential or Voltage

The definition of the potential difference between two points, a and b , in space is

$$V_b - V_a = - \int_a^b \vec{E} \cdot d\vec{l}$$

Where this integral is a *path integral*, or *line integral*.

For electric fields made by electric charges, the answer for the voltage difference is the same no matter what path is chosen.

5. Electric Potential or Voltage

Applying the previous definition to the field of a point charge, q , and choosing the convention that the voltage is zero at infinity, we obtain the formula for the voltage produced at any point in space by a point charge:

$$V = \frac{k_e q}{r}$$

Where r is the distance from the point charge to the point at which it is desired to know V .

5. Electric Potential or Voltage

This formula says that the voltage is large and positive near positive charges and large and negative near negative charges.

$$V = \frac{k_e q}{r}$$

5.1. Equipotential Surfaces

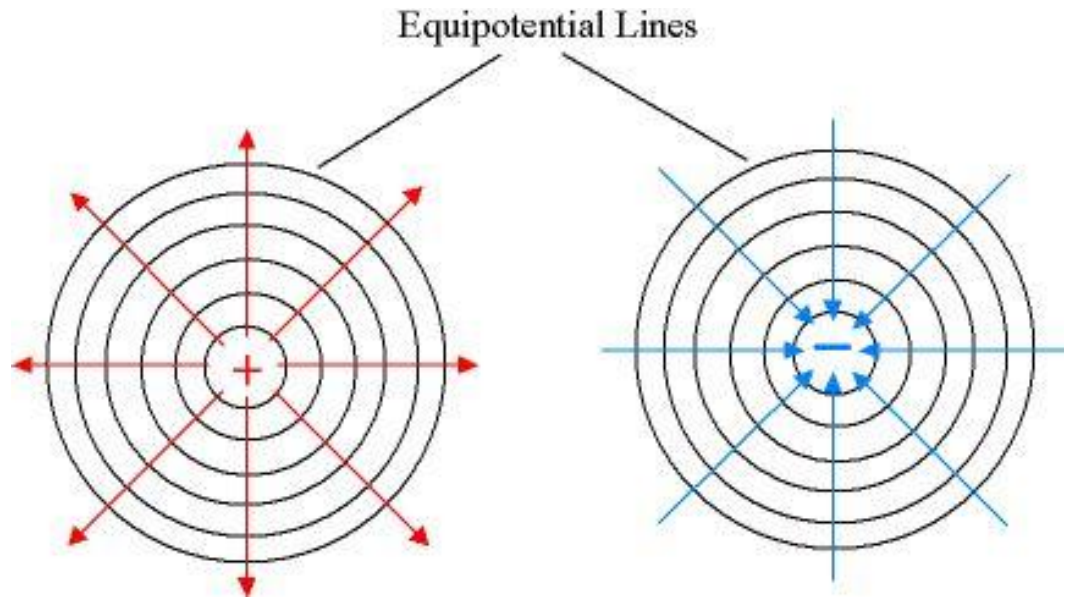
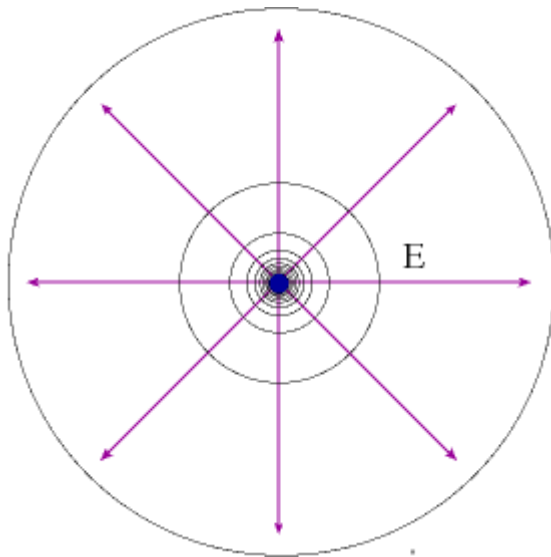
An *equipotential surface* is a surface on which the potential, or voltage, is constant.

Electric field lines are always perpendicular to these surfaces, and the electric field points from surfaces of high potential to surfaces of low potential.

Note that this means that the electric field is strong where the equipotential surfaces are close together and, weak where they are far apart.

5.1. Equipotential Surfaces

$$V(\vec{r}) = \frac{kQ}{r}$$



5.2. Potential Energy

The potential energy of a charge, q , at a point in space where the voltage *due to all charges* is V , is given by the formula

$$U = qV$$

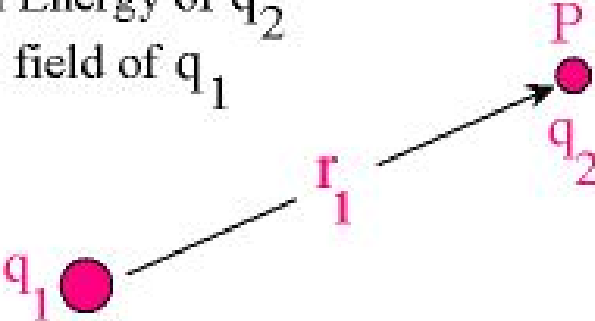
5.2. Potential Energy

for the potential energy of two point charges, q_1 and q_2 , separated by a distance r_{12} is equal to:

$$U_{12} = k_e \frac{q_1 q_2}{r_{12}}$$

P.E. = $\frac{kq_1 q_2}{r_1} = V_1 q_2$

Potential Energy of q_2
in the field of q_1



5.2. Potential Energy

The potential energy of the group is simply the sum of the potential energies for all the pairs of charges in the group.

For instance, for a group of four charges the total potential energy would be

$$U = U_{12} + U_{13} + U_{14} + U_{23} + U_{24} + U_{34}$$

An example of a group of three charges



$$U = \left[\frac{kq_1q_2}{r_{12}} + \frac{kq_2q_3}{r_{23}} + \frac{kq_1q_3}{r_{13}} \right]$$

Rearrange the expression as,

$$U = k \left[\frac{q_1q_2}{r_{12}} + \frac{q_2q_3}{r_{23}} + \frac{q_1q_3}{r_{13}} \right]$$

6. Capacitors

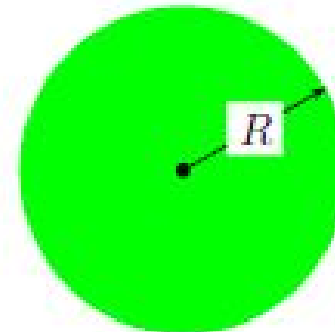
A capacitor is an arrangement of conductors that is used to store electric charge.



A very simple capacitor is an isolated metallic sphere.

The potential of a sphere with radius R and charge Q is equal to

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$



6. Capacitors

The previous equation shows that the potential of the sphere is proportional to the charge Q on the conductor.

This is true in general for any configuration of conductors.

This relationship can be written as

$$Q = CV$$

where C is called the **capacitance** of the system of conductors.

The unit of capacitance is the **farad** (F).

6. Capacitors

The capacitance of the metallic sphere is equal to

$$C = \frac{Q}{V} = \frac{Q}{\frac{1}{4\pi\epsilon_0 R} Q} = 4\pi\epsilon_0 R$$

This formula shows that the capacitance of a metallic sphere depends only on the geometry of the sphere not on the electric charges that it carries.

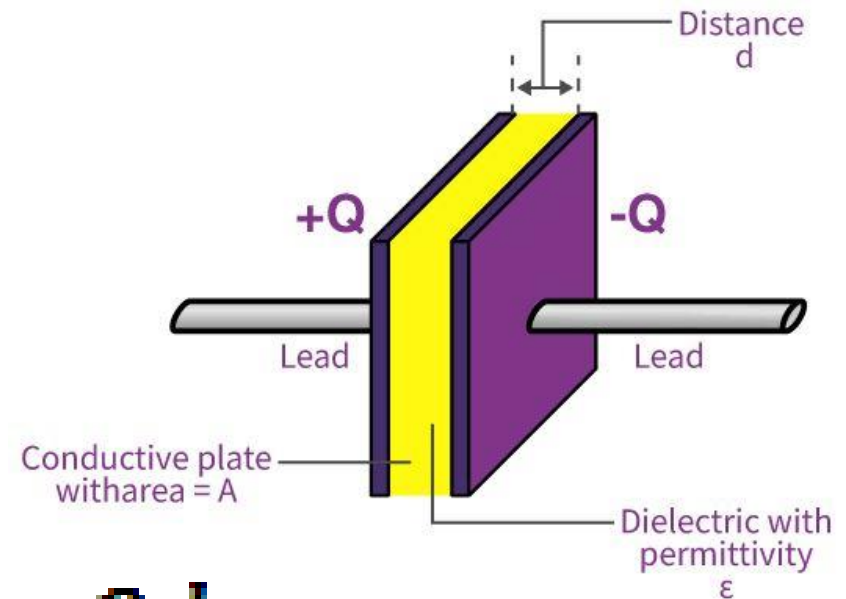
This is true in general for any configuration of conductors.

6.1. The parallel-plate capacitor

Another example of a capacitor is a system consisting of two parallel metallic plates.

The potential difference between two plates of area A , separated by a distance d , and with charges $+Q$ and $-Q$ is given by:

$$\Delta V = \frac{Qd}{\epsilon_0 A}$$



6.1. The parallel-plate capacitor

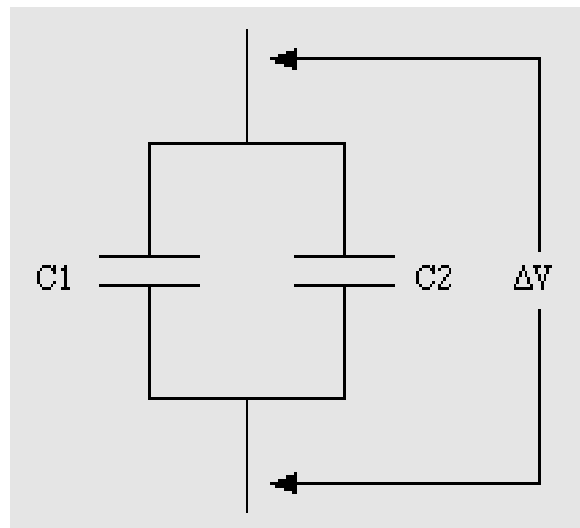
Using the definition of the capacitance, the capacitance of this system can be calculated:

$$C = \frac{Q}{\Delta V} = \frac{\epsilon_0 A}{d}$$

6.2. Capacitors in Combination

Capacitors can be connected together; they can be connected in series or in parallel.

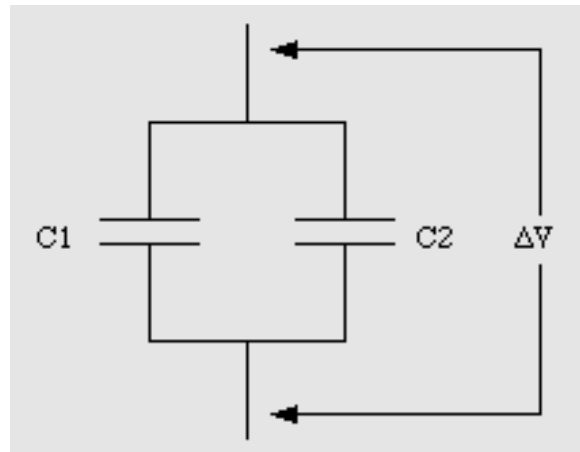
The figure below shows two capacitors, with capacitance C_1 and C_2 , connected in parallel.



6.2. Capacitors in Combination

The potential difference across both capacitors must be equal and therefore

$$\Delta V = \Delta V_1 = \frac{Q_1}{C_1} = \Delta V_2 = \frac{Q_2}{C_2}$$



6.2. Capacitors in Combination

Using equation , the total charge on both capacitors can be calculated

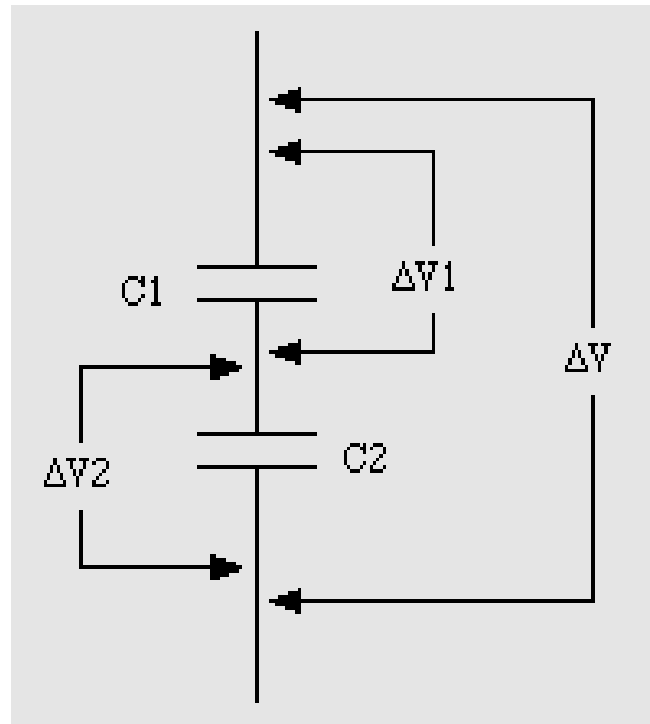
$$Q = Q_1 + Q_2 = (C_1 + C_2)\Delta V$$

These two capacitors can be treated as one capacitor with a capacitance C ; where C is related to C_1 and C_2 in the following manner

$$C = C_1 + C_2$$

6.2. Capacitors in Combination

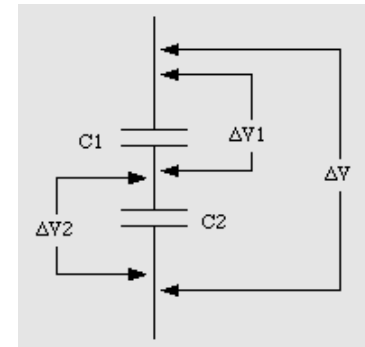
Figure below shows two capacitors, with capacitance C_1 and C_2 , connected in series.



6.2. Capacitors in Combination

- The total voltage difference across the two capacitors is given by

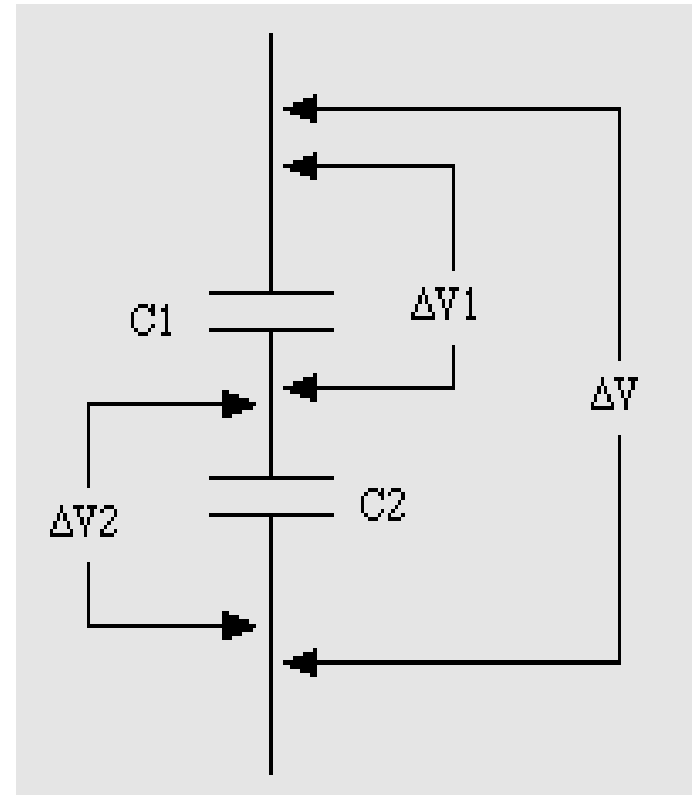
$$\Delta V = \Delta V_1 + \Delta V_2 = \left(\frac{1}{C_1} + \frac{1}{C_2} \right) Q$$



6.2. Capacitors in Combination

The system acts like a single capacitor C whose capacitance can be obtained from the following formula

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$



6.4. Energy in Capacitors

The electric potential energy of a capacitor containing no dielectric and with charge $+/-Q$ on its plates is given by

$$U = \frac{1}{2} QV_2 - \frac{1}{2} QV_1 = \frac{1}{2} Q\Delta V$$

Where V_1 and V_2 are the potentials of the two plates.

6.4. Energy in Capacitors

The electric potential energy can also be expressed in terms of the capacitance C of the capacitor

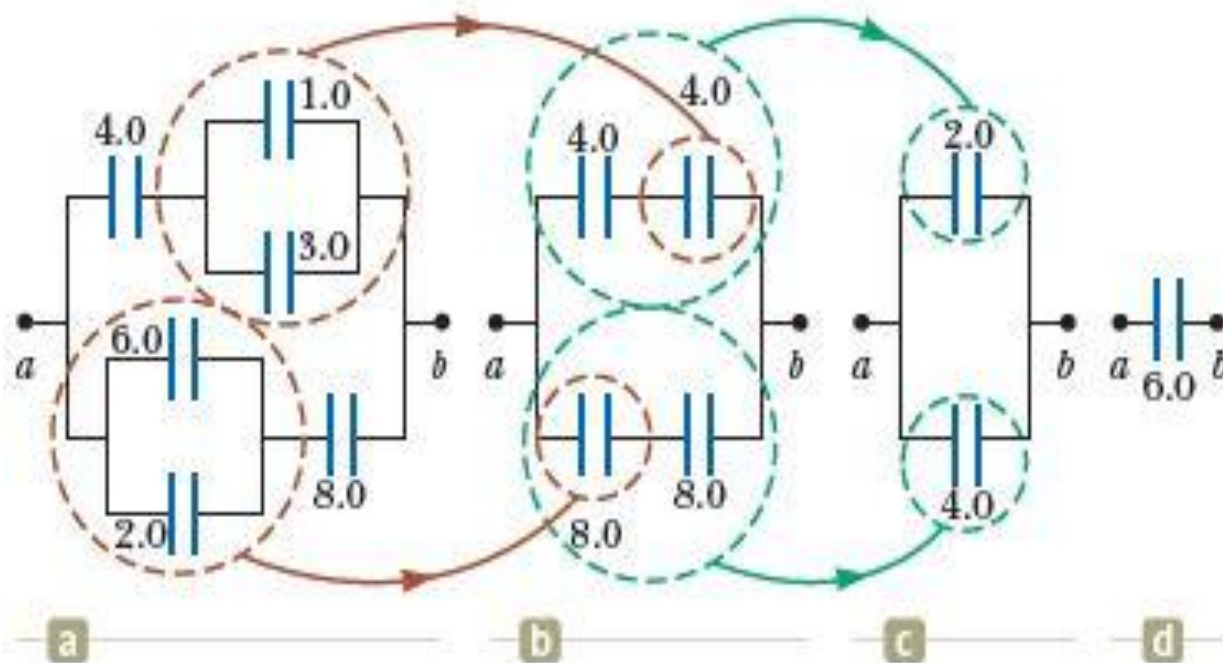
$$U = \frac{1}{2} Q \frac{Q}{C} = \frac{Q^2}{2C}$$

This formula is also correct for a capacitor with a dielectric; the properties of the dielectric enter into this formula via the capacitance C .

Exercise 9

Find the equivalent capacitance between *a* and *b* for the combination of capacitors shown in Figure below. All capacitances are in microfarads.

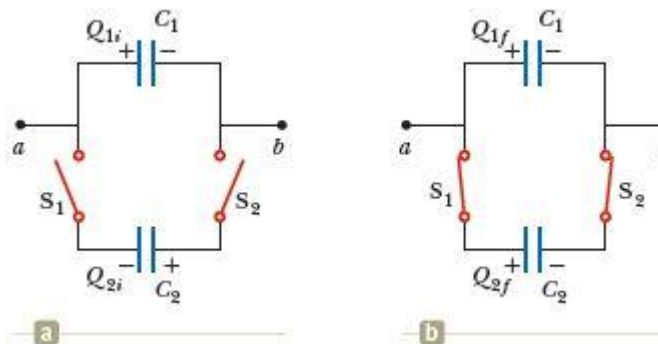
Solution



Exercise 10

Two capacitors C_1 and C_2 (where $C_1 > C_2$) are charged to the same initial potential difference Δv_i . The charged capacitors are removed from the battery, and their plates are connected with opposite polarity as in Figure a. The switches S_1 and S_2 are then closed as in Figure b.

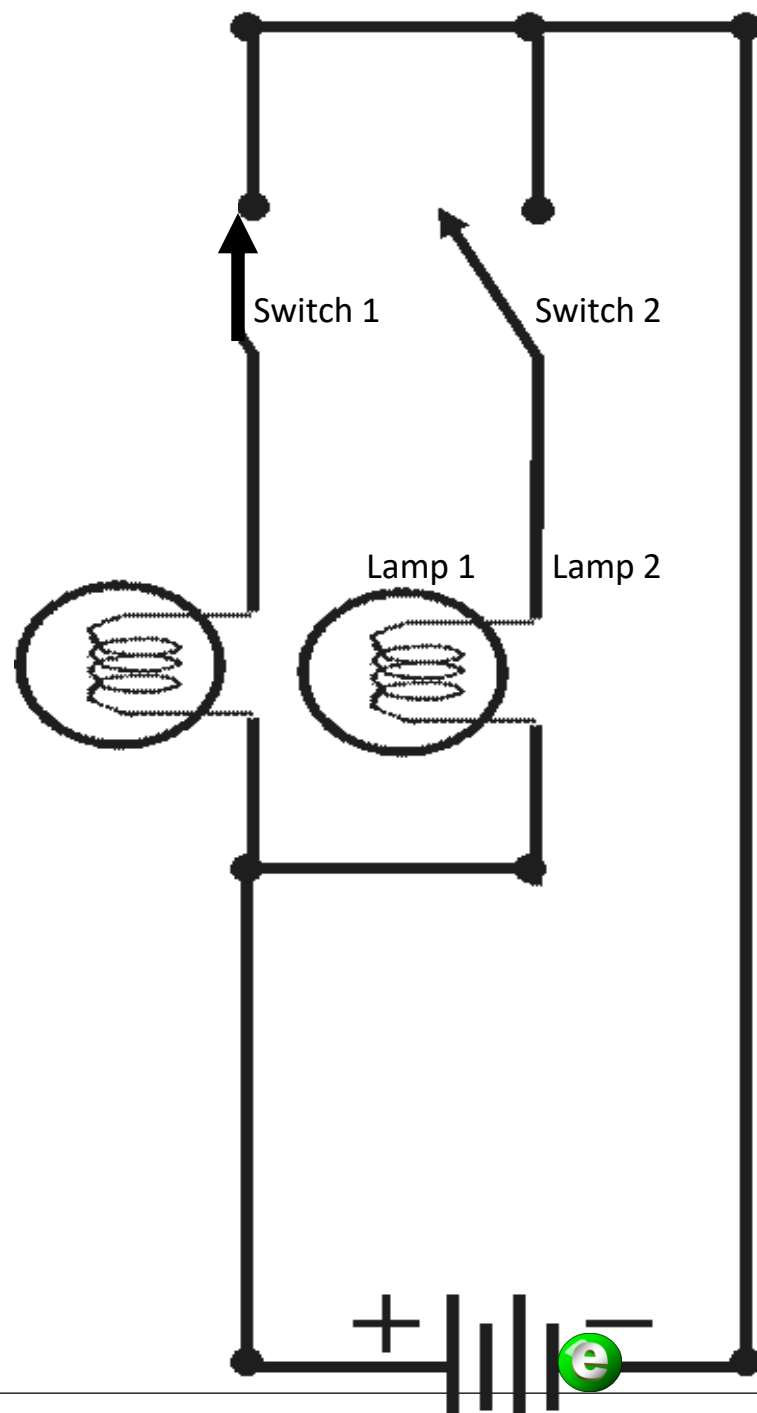
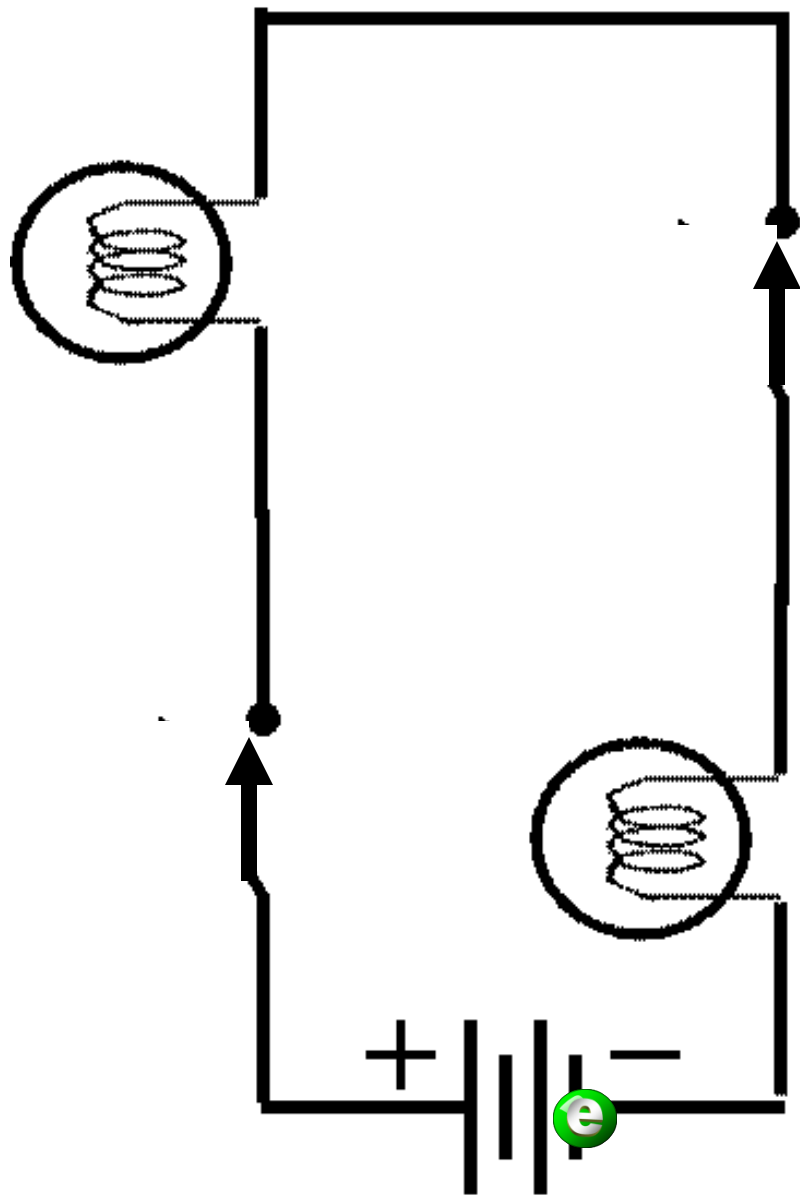
- Find the final potential difference Δv_f between a and b after the switches are closed
- Find the total energy stored in the capacitors before and after the switches are closed and determine the ratio of the final energy to the initial energy.



7. Electric current

The movement of electric charges is known as an electric current, the intensity of which is usually measured in **amperes (A)**.

Electric current can consist of any moving charged particles; most commonly these are electrons, but any charge in motion constitutes an electric current.



7. Electric current

By historical convention, a positive electric current is defined as having the same direction of flow as any positive charge it contains, or to flow from the most positive part of a circuit to the most negative part.

Electric current defined in this manner is called “*conventional electric current*.”

7. Electric current

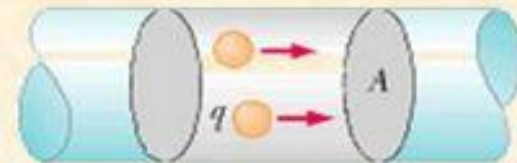
However, depending on the conditions, an electric current can consist of a flow of charged particles in either direction or even in both directions at once.

The positive-to-negative convention is widely used to simplify this situation.

7. Electric current

- An **electric current** I is a measure of the **rate of flow of electric charge** Q through a given cross section of a conductor.
- **Symbol** of Electric Current = I
- **SI Unit** of Electric Current = **ampere (A)**

$$I = Q/t$$



where

I = current in ampere (A)

Q = amount of charges in coulombs (C)

t = time in seconds (s)

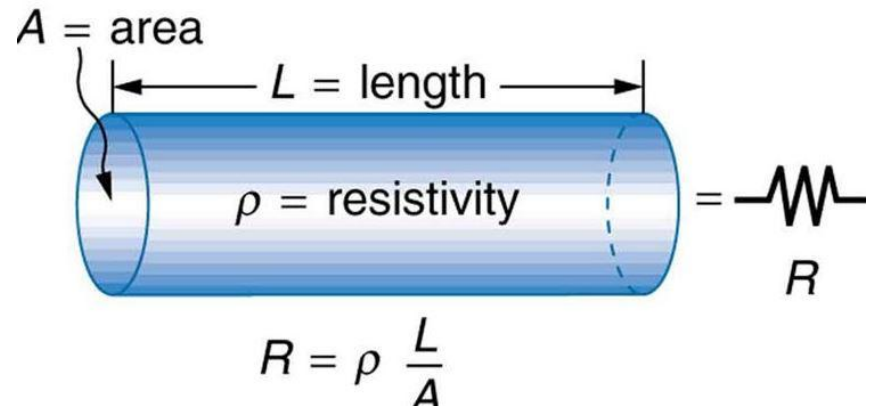
7.1. Electrical resistance

Voltage can be thought of as the pressure pushing charges along a conductor, while “*the electrical resistance of a conductor is a measure of how difficult it is to push the charges along*”.

Using the water flow analogy, electrical resistance is similar to friction.

7.1. Electrical resistance

The resistance, R , of a material depends on its length (L), cross-sectional area (A), and the resistivity (ρ), a number that depends on the material:



$$R = \frac{\rho L}{A}$$

ρ = resistivity
 L = length
 A = cross sectional area

7.1. Electrical resistance

Resistance also depends on temperature, usually increasing as the temperature increases the resistance.

$$R = R_0(1 + \alpha \Delta T)$$

- R_0 : reference resistance usually at 20°C (sometimes at 0° C)

of the form

$$\rho = \rho_0 [1 + \alpha (T - T_0)],$$

where ρ_0 is the resistivity at temperature T_0 , and α is the temperature coefficient of resistivity.

7.1. Electrical resistance

At low temperatures some materials, known as superconductors, have no resistance at all.

Resistance in wires produces a loss of energy (usually in the form of heat), so materials with no resistance produce no energy loss when currents pass through them.

7.2. Ohm's Law

For a current to exist in a conductor, there must be a potential difference between its ends, just as a difference in height between source and outlet is necessary for a river current to exist.

In the case of a metallic conductor, the current is proportional to the applied potential difference: *Doubling V causes I to double, tripling V causes I to triple, and so forth.*

7.2. Ohm's Law

Given:

V = Voltage

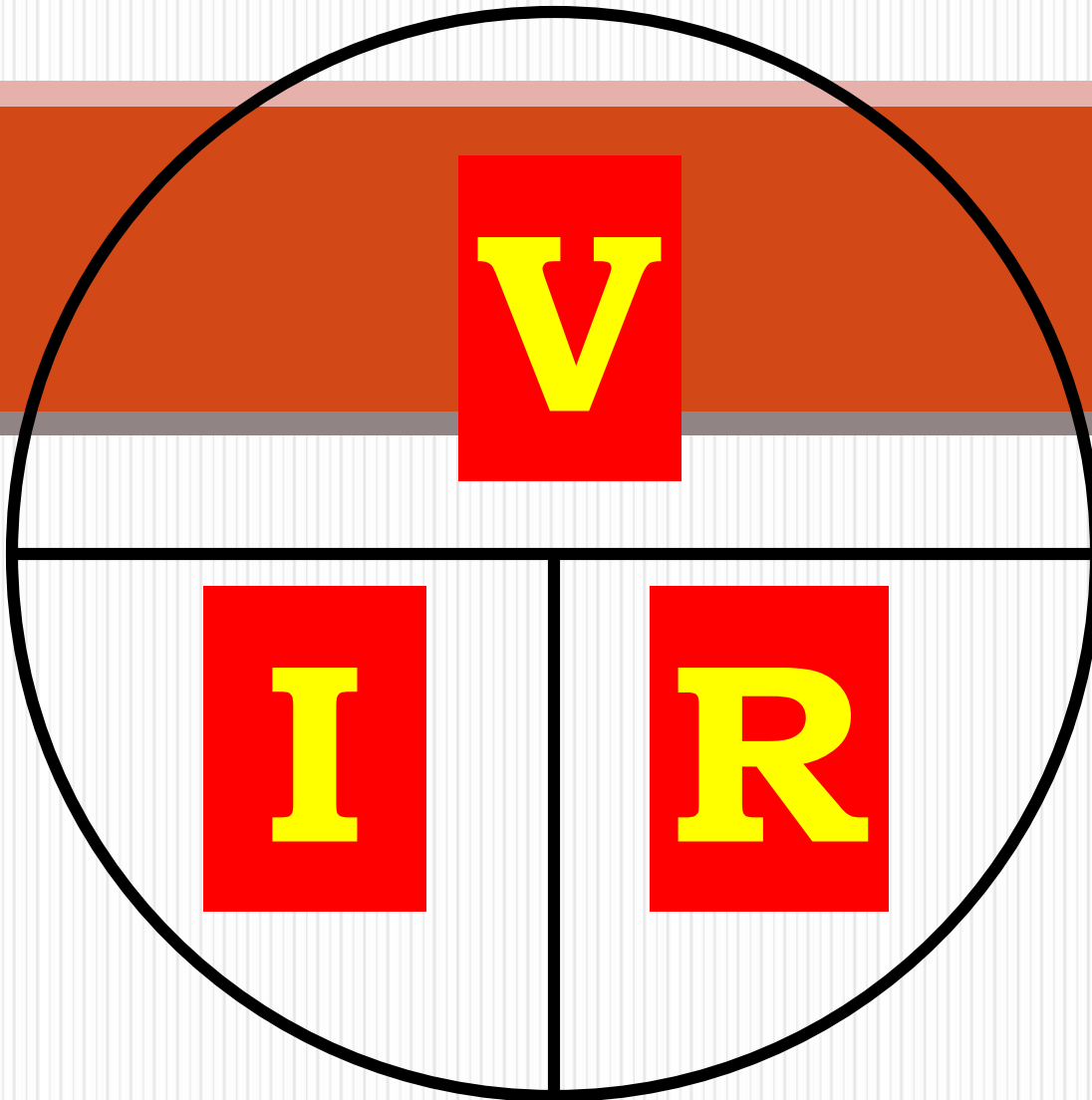
I = Current

R = Resistance

$$\mathbf{V=IR}$$

$$\mathbf{I = V/R}$$

$$\mathbf{R = V/I}$$



7.4. Electromotive force and internal resistance

Let us consider a battery as a source of energy for circuits. Because the potential difference at the battery terminals is constant in a particular circuit, the current in the circuit is constant in magnitude and direction and is called **direct current**. A battery is called either a *source of electromotive force* or, more commonly, a *source of emf*. (The phrase *electromotive force* is an unfortunate historical term, describing not a force, but rather a potential difference in volts.)

7.4. Electromotive force and internal resistance

The **emf** $\mathcal{E}$ of a battery is the maximum possible voltage the battery can provide between its terminals (*i.e. the **emf** is equal to the potential difference across the terminals of the source when no current flows.*) You can think of a source of emf as a “charge pump.”

When an electric potential difference exists between two points, the source moves charges “uphill” from the lower potential to the higher. We shall generally assume the connecting wires in a circuit have no resistance. The positive terminal of a battery is at a higher potential than the negative terminal.

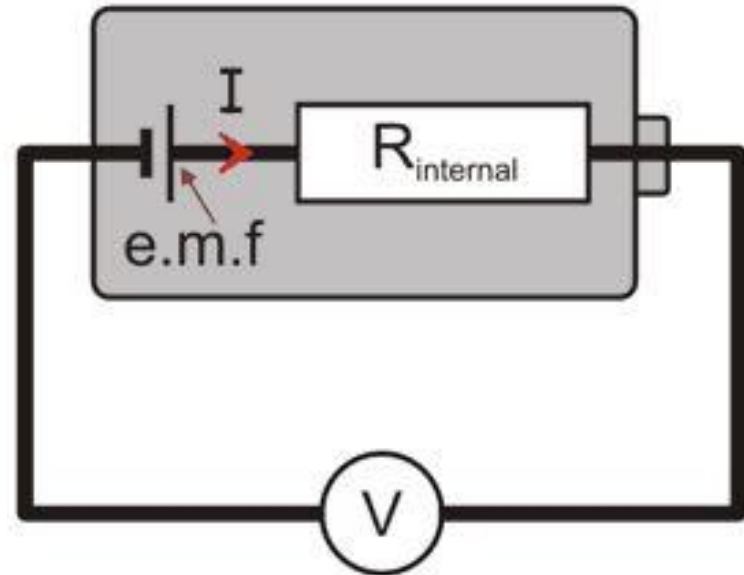
7.4. Electromotive force and internal resistance

The emf, internal resistance, terminal voltage and current drawn can be linked by the following equation:

$$\varepsilon = V + Ir$$

where

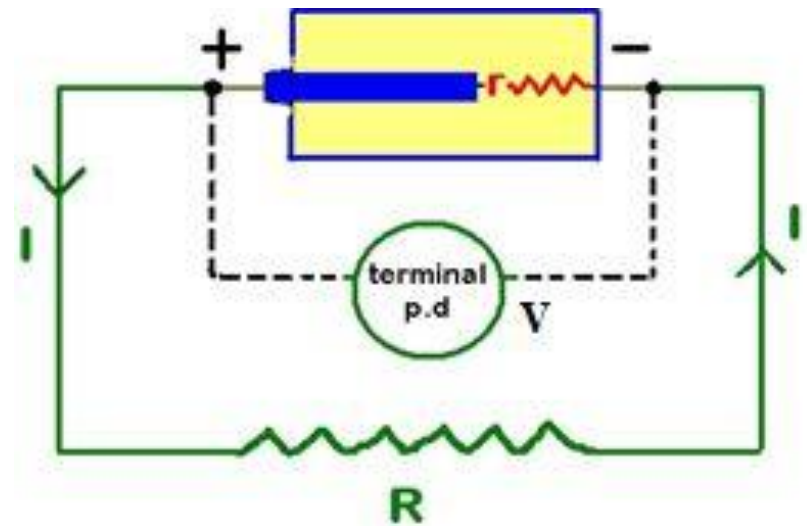
***r** is the internal resistance and
V the terminal voltage*



7.4. Electromotive force and internal resistance

When a battery or generator of emf $\mathcal{E}$ is connected to an external resistance R , the total resistance in the circuit is $R + r$, and the current that flows is

$$I = \frac{\mathcal{E}}{R + r}$$



example

Q1) The p.d. across the terminals of a cell is 3.0 volts when it is not connected to a circuit and no current is flowing. When the cell is connected to a circuit and a current of 0.37 A is flowing the terminal p.d. falls to 2.8 V. What is the internal resistance of the cell?

$$Q1) \quad \mathcal{E} = V + Ir$$

$$\mathcal{E} - V = Ir$$

$$\frac{\mathcal{E} - V}{I} = r$$

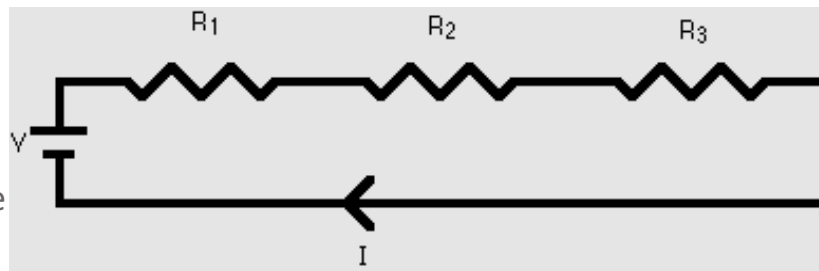
$$r = \frac{3.0 - 2.8}{0.37} = \underline{\underline{0.54 \, \Omega}}$$

7.5. Series and parallel circuits

a) Series circuits

A series circuit is a circuit in which resistors are arranged in a chain, so the current has only one path to take.

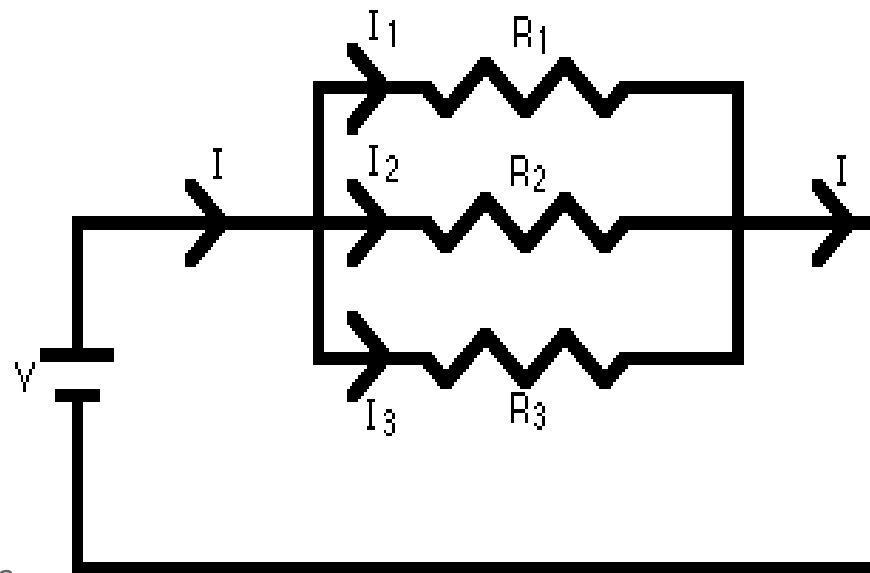
The current is the same through each resistor. The total resistance of the circuit is found by simply adding up the resistance values of the individual resistors. $R = R_1 + R_2 + R_3 + \dots$



7.5. Series and parallel circuits

b) Parallel circuits

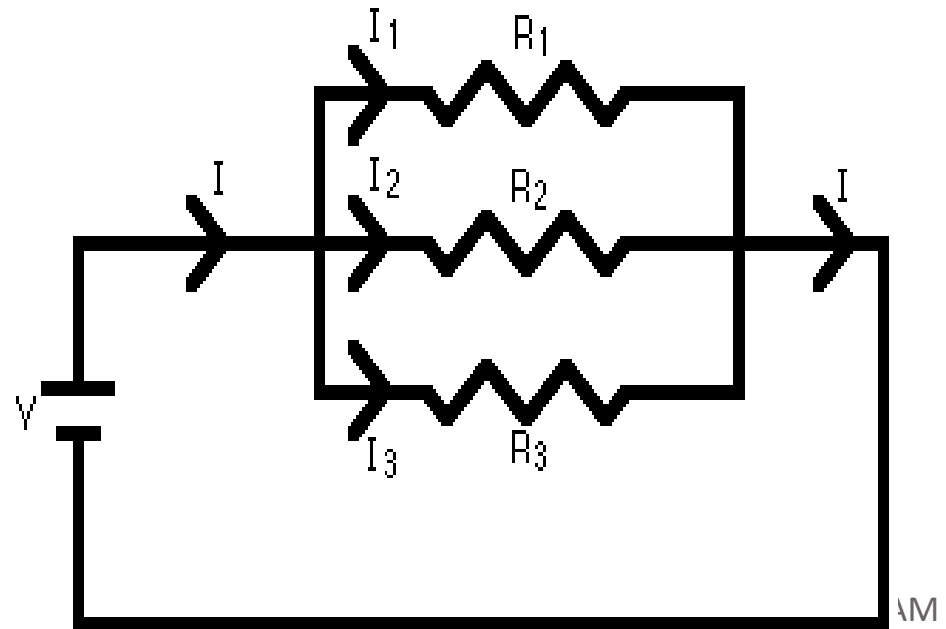
A parallel circuit is a circuit in which the resistors are arranged with their heads connected together, and their tails connected together.



7.4. Series and parallel circuits

The current in a parallel circuit breaks up, with some flowing along each parallel branch and re-combining when the branches meet again.

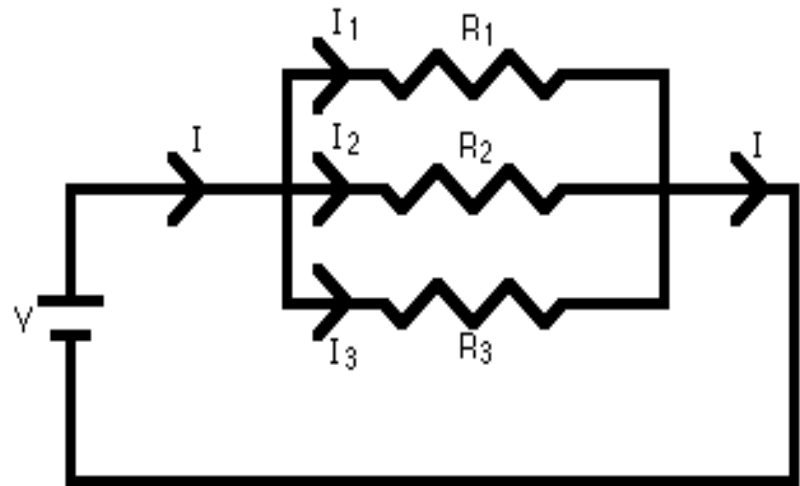
The voltage across each resistor in parallel is the same.

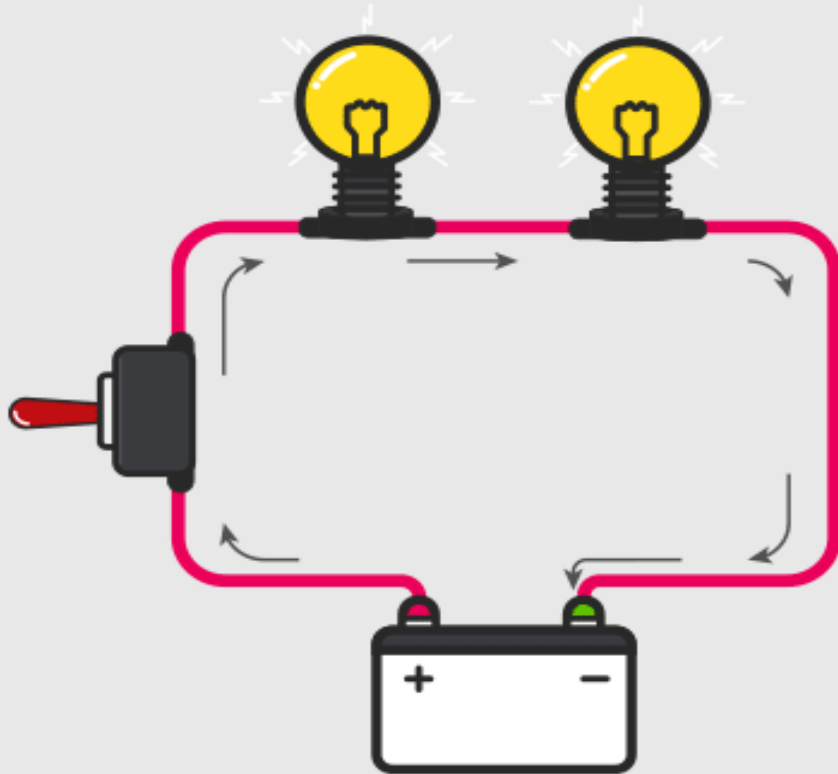


7.4. Series and parallel circuits

The total resistance of a set of resistors in parallel is found by adding up the reciprocals of the resistance values, and then taking the reciprocal of the total:

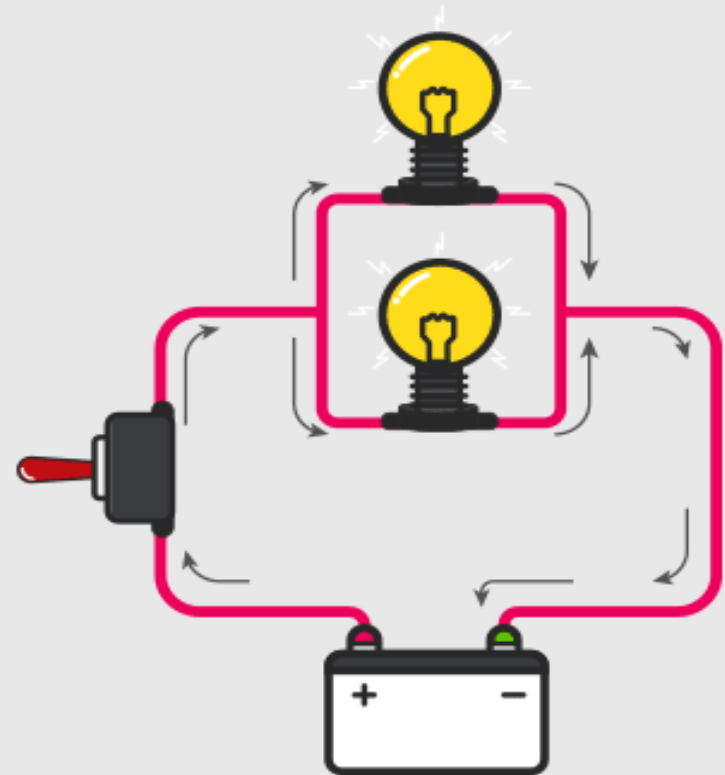
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$





SERIES CIRCUITS

A SERIES CIRCUIT IS MADE BY CONNECTING THE END OF ONE DEVICE TO THE BEGINNING OF ANOTHER



PARALLEL CIRCUITS

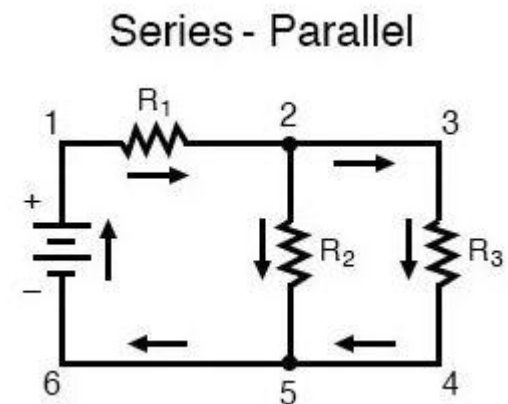
IN PARALLEL CIRCUITS THE SAME TERMINALS OF BOTH DEVICES ARE CONNECTED TOGETHER

7.4. Series and parallel circuits

c) Circuits with series and parallel components

Many circuits have a combination of series and parallel resistors.

Generally, the total resistance in a circuit like this is found by reducing the different series and parallel combinations step-by-step to end up with a single equivalent resistance for the circuit. This allows the current to be determined easily.



Quiz

Exercise 11

The radius of 22-gauge Nichrome wire is **0.32 mm** and its resistivity is **$1.0 \times 10^{-6} \Omega/m$**

- (A) Calculate the resistance per unit length of this wire.
- (B) If a potential difference of 10 V is maintained across a **1.0 m** length of the Nichrome wire, what is the current in the wire?

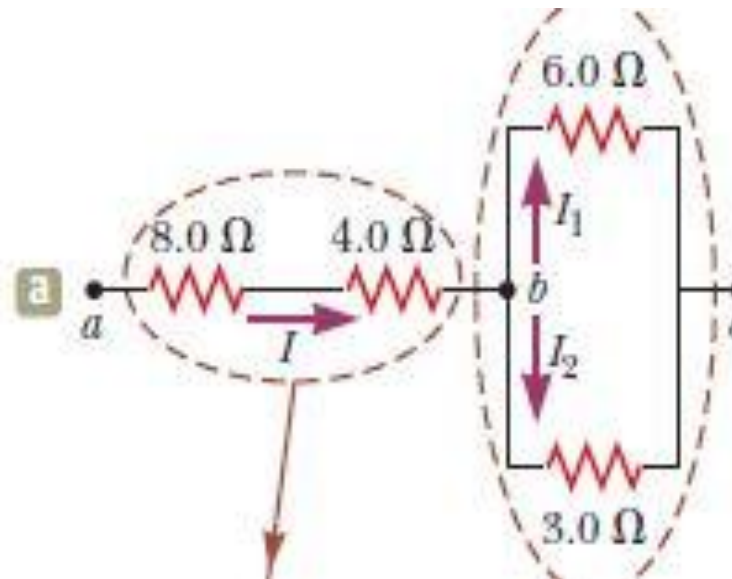
Exercise 12

A battery has an emf of **12.0 V** and an internal resistance of **0.0500Ω** . Its terminals are connected to a load resistance of **3.00Ω** . Find the current in the circuit and the terminal voltage of the battery.

Exercise 13

Four resistors are connected as shown in Figure below

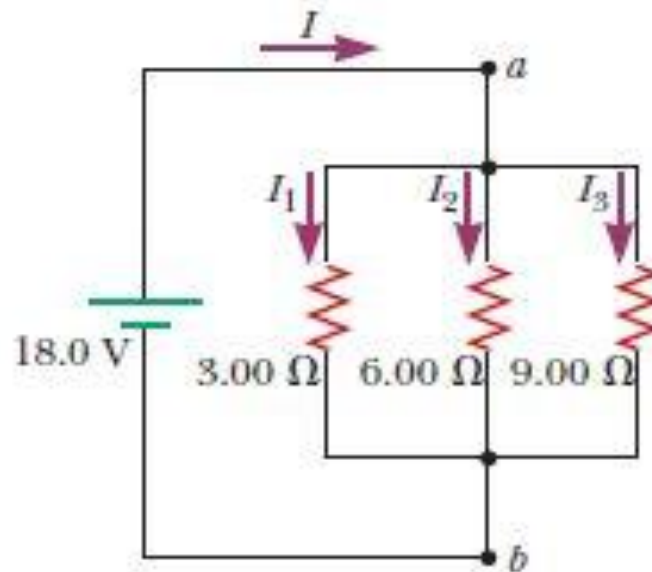
- (A) Find the equivalent resistance between points a and c .
- (B) What is the current in each resistor if a potential difference of 42 V is maintained between a and c ?



Exercise 14

Three resistors are connected in parallel as shown in Figure below. A potential difference of 18.0 V is maintained between points a and b .

- (A) Calculate the equivalent resistance of the circuit.
- (B) Find the current in each resistor.



8. Electrical Energy and power

a) Electric energy

As already mentioned, voltage is energy per unit charge, so if you want to know how much energy has been provided by a battery, you simply multiply the voltage by the amount of charge that it has moved so far.

8. Electrical Energy and power

The same formula gives the amount of electrical energy *used* by a light bulb or any other device that consumes electrical energy, provided that you replace the battery voltage by the potential difference (voltage *difference*) between one end of the device and the other:

$$U = V \times q$$

8. Electrical Energy and power

In the case of resistive (Ohmic, or linear) loads; the definition of electrical current, combined with Ohm's law ($V = RI$) the expression for the energy is given by

$$U = RI^2t$$

The unit of electric energy is watt-second (Ws) but in commercial domain they prefer to use kilowatt-hour (kWh)

8. Electrical Energy and power

a) Electric power

Electric power is the rate at which electric energy is transferred by an electric circuit.

The SI unit of power is the **watt**.

Electric power, like mechanical power, is represented by the letter ***P*** in electrical equations.

The term *wattage* is used colloquially to mean "electric power in watts."

8. Electrical Energy and power

In direct current resistive circuits, electrical power is calculated using Joule's law:

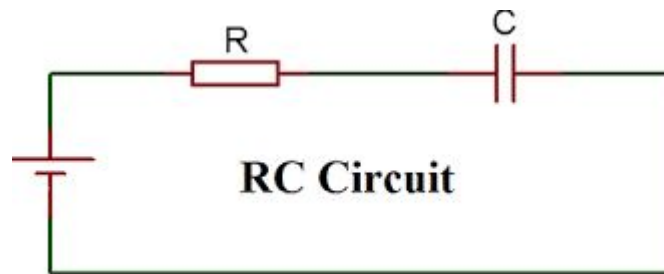
$$P = VI$$

In the case of resistive (Ohmic, or linear) loads, Joule's law can be combined with Ohm's law ($I = V/R$) to produce alternative expressions for the dissipated power:

$$P = I^2 R = \frac{V^2}{R}$$

RC Circuits

We have analyzed direct-current circuits in which the current is constant. In Direct Circuits (**DC circuits**) containing capacitors, the current is always in the same direction but may vary in magnitude at different times. A circuit containing a series combination of a resistor and a capacitor is called an *RC circuit*.



Charging a capacitor

while switch S is open (Figure 1.38b). If the switch is closed at $t = 0$, however, charge begins to flow, setting up a current in the circuit, and the capacitor begins to charge. Note that during charging, charges do not jump across the capacitor plates because the gap between the plates represents an open circuit. Instead, charge is transferred between each plate and its connecting wire due to the electric field established in the wires by the battery, until the capacitor is fully charged. As the plates become charged, the potential difference across the capacitor increases. The value of the maximum charge depends on the voltage of the battery. Once the maximum charge is reached, the current in the circuit is zero because the potential difference across the capacitor matches that supplied by the battery.

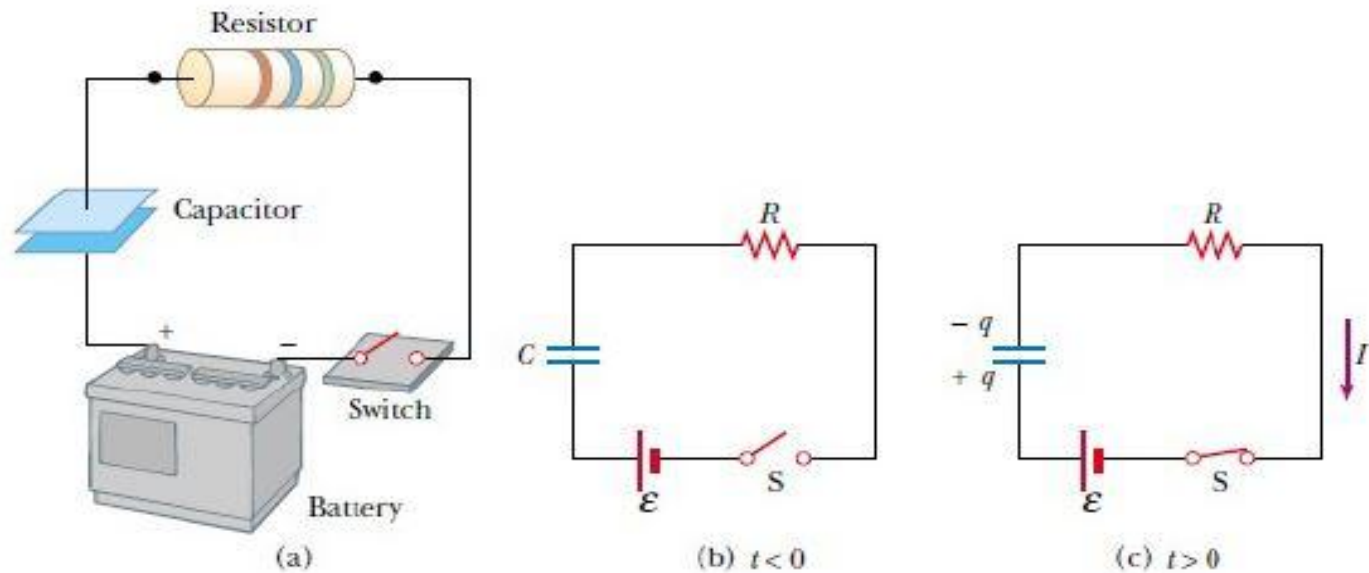


Figure 1.38: (a) A capacitor in series with a resistor, switch, and battery. (b) Circuit diagram representing this system at time $t < 0$, before the switch is closed. (c) Circuit diagram at time $t > 0$, after the switch has been closed.

Charging a capacitor

To analyze this circuit quantitatively, let us apply Kirchhoff's loop rule to the circuit after the switch is closed. Traversing the loop clockwise gives

$$\mathcal{E} - \frac{q}{C} - IR = 0 \quad (1.4.18)$$

where q/C is the potential difference across the capacitor and IR is the potential difference across the resistor.

At the instant the switch is closed ($t = 0$), the charge on the capacitor is zero, and from the previous equation we find that the initial current in the circuit I_0 is a maximum and is equal to

$$I_0 = \frac{\mathcal{E}}{R} \quad (1.4.19)$$

At this time ($t = 0$), the potential difference from the battery terminals appears entirely across the resistor. Later, when the capacitor is charged to its maximum value Q , charges cease to flow, the current in the circuit is zero, and the potential difference from the battery terminals appears entirely across the capacitor. Substituting $I = 0$ into the Equation 1.4.18 gives the charge on the capacitor at this time:

$$Q = C\mathcal{E} \quad (1.4.20)$$

Charging a capacitor

To determine analytical expression for the time dependence of the charge and current, we must solve the equation Equation 1.4.18. The current in all parts of series circuit must be the same. Thus, the current in the resistance R must be the same as the current flowing out of and into the capacitor plates. This current is equal to the time rate of change of the charge on the capacitor plates. Thus, we substitute $I = dq/dt$ into Equation 1.4.18 and rearrange the equation to get

$$\frac{dq}{dt} = -\frac{q - C\mathcal{E}}{RC} \quad (1.4.21)$$

or

$$\frac{dq}{q - C\mathcal{E}} = -\frac{1}{RC} dt \quad (1.4.22)$$

Integrating this expression, using the fact that $q = 0$ at $t = 0$, we obtain

$$\int_0^q \frac{dq}{q - C\mathcal{E}} = -\frac{1}{RC} \int_0^t dt \quad (1.4.23)$$

That is

$$\ln \left(\frac{q - C\mathcal{E}}{-C\mathcal{E}} \right) = -\frac{t}{RC} \quad (1.4.24)$$

Charging a capacitor

From the definition of the natural logarithm, we can write this expression as

$$q(t) = C\mathcal{E} (1 - e^{-t/RC}) = Q (1 - e^{-t/RC}) \quad (1.4.25)$$

We can find an expression for the charging current by differentiating the previous equation with respect to time. Using $I = dq/dt$ we find that

$$I(t) = \frac{\mathcal{E}}{R} e^{-t/RC} \quad (1.4.26)$$

The quantity RC is called the *time constant* τ of the circuit. The time constant represents the time interval during which the current decreases to $1/e$ of its initial value; that is, after a time interval τ , the current decreases to $i = e^{-1}I_i = 0.368I_i$. After a time interval 2τ , the current decreases to $i = e^{-2}I_i = 0.135I_i$ and so forth. Likewise, in the time interval τ , the charge increases from zero to $C\mathcal{E}[1 - e^{-1}] = 0.632C\mathcal{E}$.

The dimensional analysis shows that τ has units of time.

The energy output of the battery as the capacitor is fully charged is $Q\mathcal{E} = C\mathcal{E}^2$. After the capacitor is fully charged, the energy stored in the capacitor is $\frac{1}{2}Q\mathcal{E} = \frac{1}{2}C\mathcal{E}^2$, which is just half the energy output of the battery. The remaining half of the energy supplied by the battery appears as internal energy in the resistor.

Discharging a Capacitor

Now let us consider the circuit shown in Figure 1.39, which consists of a capacitor carrying an initial charge Q , a resistor, and a switch. The initial charge Q is not the same as the maximum charge Q in the previous discussion, unless the discharge occurs after the capacitor is fully charged (as described earlier). When the switch is open, a potential difference Q/C exists across the capacitor and there is zero potential difference across the resistor because $I = 0$. If the switch is closed at $t = 0$ the capacitor begins to discharge through the resistor. At some time t during the discharge, the current in the circuit is I and the charge on the capacitor is q (Figure 1.39b). The circuit in Figure 1.39 is the same as the circuit in Figure 1.38 except for the absence of the battery. Thus, we eliminate the *emf* from the Equation 1.4.18 to obtain the appropriate loop equation for the circuit in Figure 1.39:

$$-\frac{q}{C} - RC = 0 \quad (1.4.27)$$

When we substitute $I = dq/dt$ into Equation 1.4.27, it becomes

$$-R \frac{dq}{dt} = \frac{q}{C} \quad \text{or} \quad \frac{dq}{q} = -\frac{1}{RC} dt \quad (1.4.28)$$

Integrating Equation 1.4.28, using the fact that $q = Q$ at $t = 0$, gives

$$\int_Q^q \frac{dq}{q} = -\frac{1}{RC} \int_0^t dt \quad (1.4.29)$$

Thus,

$$\ln \left(\frac{q}{Q} \right) = -\frac{t}{RC}$$

or

$$q(t) = Qe^{-t/RC} \quad (1.4.30)$$

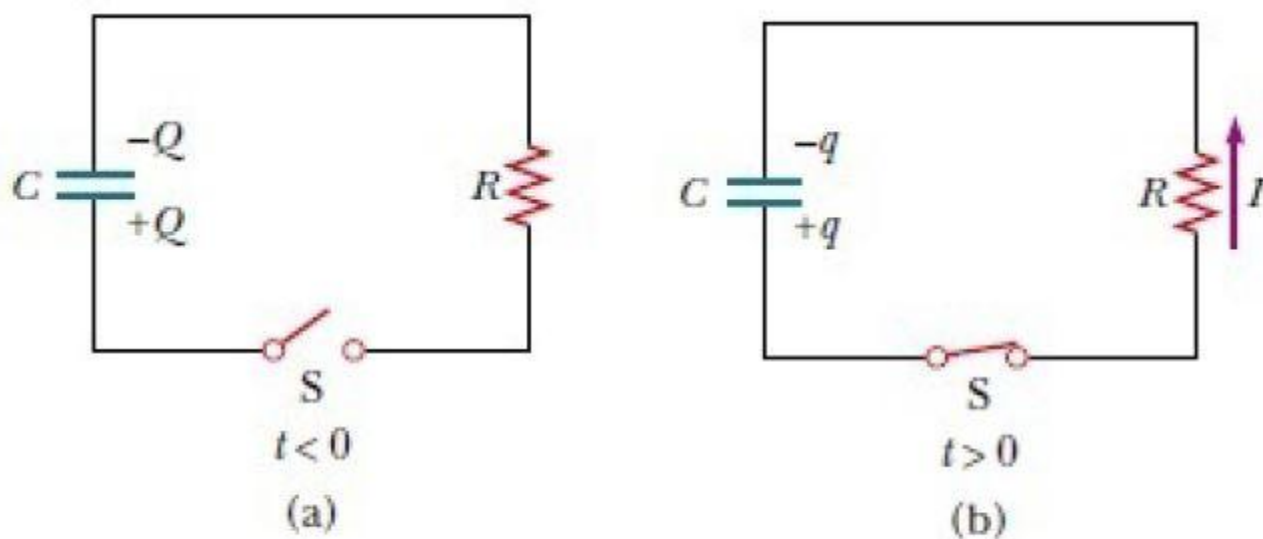


Figure 1.39: (a) A charged capacitor connected to a resistor and a switch, which is open at $t < 0$ (b) After the switch is closed, a current that decreases in magnitude with time is set up in the direction shown, and the charge on the capacitor decreases exponentially with time.

Differentiating the expression of the Equation 1.4.30 with respect to time gives the instantaneous current as a function of time:

$$I(t) = \frac{dq}{dt} = \frac{d}{dt} (Qe^{-t/RC}) = -\frac{Q}{RC}e^{-t/RC} \quad (1.4.31)$$

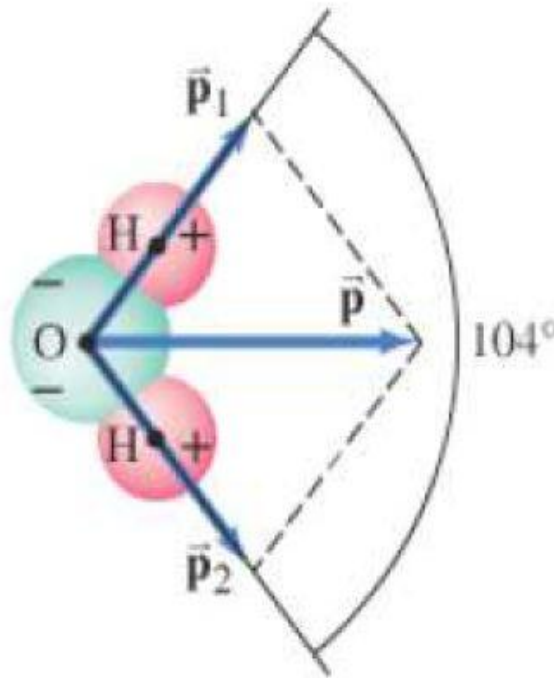
where $Q/RC = I_0$ is the initial current. The negative sign indicates that the current direction now that the capacitor is discharging is opposite the current direction when the capacitor was being charged. We see that both the charge on the capacitor and the current decay exponentially at a rate characterized by the time constant $\tau = RC$.

Exercises

1. A $+35 \mu\text{C}$ point charge is placed 32 cm from an identical $+35 \mu\text{C}$ charge. How much work would be required to move a $+0.50 \mu\text{C}$ test charge from a point midway between them to a point 12 cm close to either of the charges?
2. In the Bohr model of the hydrogen atom, an electron orbits a proton (the nucleus) in a circular orbit of radius $0.53 \times 10^{-10} \text{ m}$. (a) What is the electric potential at the electron's orbit due to the proton? (b) What is the kinetic energy of the electron? (c) What is the total energy of the electron in its orbit? (d) What is the ionization energy—that is, the energy required to remove the electron from the atom and take it to $r = \infty$, at rest? Express the results of parts b, c and d in joules and eV.

Exercises

3. The dipole moment, considered as a vector, points from the negative to the positive charge. The water molecule, has a dipole moment $\vec{P}$ which can be considered as the vector sum of the two dipole moments, $\vec{P}_1$ and $\vec{P}_2$, as shown in the figure. The distance between each H and the O is about $0.96 \times 10^{-10} \text{ m}$. The lines joining the center of the O atom with each H atom make an angle of 104° , as shown, and the net dipole moment has been measured to be $p = 6.1 \times 10^{-30} \text{ C} \cdot \text{m}$. Determine the charge q on each H atom.



Exercises

4. A $2.50 - \mu F$ capacitor is charged to 857 V and a $6.80 - \mu F$ capacitor is charged to 652 V. These capacitors are then disconnected to each other and the negative plates are connected to each other. What will be the potential difference across each and the charge on each?
5. A homemade capacitor is assembled by placing two 9-in. pie pans 5 cm apart and connecting them to the opposite terminals of a 9-V battery. Estimate (a) the capacitance, (b) the charge on each plate, (c) the electric field halfway between the plates, and (d) the work done by the battery to charge the plates. (e) Which of the above values change if a dielectric is inserted?
6. A cardiac defibrillator is used to shock a heart that is beating erratically. A capacitor in this device is charged to 5.0 kV and stores 1200 J of energy. What is its capacitance?
7. A $2.70 - \mu F$ capacitor is charged by a 12.0-V battery. It is disconnected from the battery and then connected to an uncharged $4.00 - \mu F$ capacitor. Determine the total stored energy (a) before the two capacitors are connected, and (b) after they are connected. (c) What is the change in energy?
8. An electron is accelerated horizontally from rest in a television picture tube by a potential difference of 5500 V. It then passes between two horizontal plates 6.5 cm long and 1.3 cm apart that have a potential difference of 250 V. At what angle θ will the electron be traveling after it passes between the plates?

- Find the current through each battery. / 10marks

